

Site-saturation mutagenesis of 500 human protein domains

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Missense variants that change the amino acid sequences of proteins cause one-third of human genetic diseases¹. Tens of millions of missense variants exist in the current human population, and the vast majority of these have unknown functional consequences. Here we present a large-scale experimental analysis of human missense variants across many different proteins. Using DNA synthesis and cellular selection experiments we quantify the effect of more than 500,000 variants on the abundance of more than 500 human protein domains. This dataset reveals that 60% of pathogenic missense variants reduce protein stability. The contribution of stability to protein fitness varies across proteins and diseases and is particularly important in recessive disorders. We combine stability measurements with protein language models to annotate functional sites across proteins. Mutational effects on stability are largely conserved in homologous domains, enabling accurate stability prediction across entire protein families using energy models. Our data demonstrate the feasibility of assaying human protein variants at scale and provides a large consistent reference dataset for clinical variant interpretation and training and benchmarking of computational methods.

The human genome encodes more than 20,000 proteins. Missense variants in nearly 5,000 of these proteins cause Mendelian diseases², however the functional consequences of nearly all missense variants in nearly all proteins are unknown^{3–5}. Given the current size of the human population, most variants compatible with life are present in someone currently alive^{6–8}, making the large-scale experimental analysis of variant function a central challenge for human genetics^{7–9}. However, experiments have so far mostly quantified variant effects in one or a few proteins^{7,10}. Despite recent improvements^{11–14}, computational variant effect predictors (VEPs) are not deemed to provide sufficient evidence to classify clinical variants as pathogenic or benign¹⁵. They also do not identify the molecular mechanisms by which variants cause disease, information that is important for therapy development and clinical trial design. Whereas many disease variants are likely to destabilize proteins and reduce their abundance^{8,16,17}, others may affect specific molecular interactions or cause gain-of-function phenotypes^{18,19}.

Previous studies have established reduced abundance as a frequent causal mechanism for pathogenic variants in diverse proteins^{8,20–24}, but larger-scale studies of human disease variants across many disease genes to test the generality of these observations are lacking. Most human proteins contain multiple independently folding structural units called domains^{25,26}. For example, the human genome encodes more than 200 homeodomains that bind DNA to control gene expression and more than 250 PDZ domains that mediate protein–protein interactions^{27,28}. The small size of protein domains (median around 100 amino acids) and their independent folding make them a useful target for large-scale experimental measurement of variant effects²⁹.

Here, using a highly validated assay that quantifies the effects of variants on protein abundance in cells³⁰, we perform large-scale mutagenesis of human protein domains. We report the effect of more than 500,000 missense variants on the stability of more than 500 different human domains. This dataset, ‘Human Domainome 1’, provides a large reference dataset for the interpretation of clinical genetic variants and for benchmarking and training computational methods for prediction of variant effects on stability. We use the dataset to quantify the contribution of stability changes to human genetic disease and how this varies across proteins and diseases. We also show how stability measurements can be combined with protein language models to annotate functional sites across proteins, and that measurements made on a small number of proteins can be used to accurately predict stability changes across entire protein families.

Mutagenesis of human protein domains

We used microchip-based massive in parallel synthesis (mMPS) technology³¹ to construct a library of 1,230,584 amino acid variants in 1,248 structurally diverse protein domains (Supplementary Table 1). In this Human Domainome 1 library, every amino acid is mutated to all other 19 amino acids at every position in each domain (Fig. 1a). Sequencing the library revealed it to be of high quality, with coverage of 91% of designed amino acid substitutions (Supplementary Table 1, Extended Data Fig. 1a). To quantify the effect of these variants on domain stability, we used an abundance protein fragment complementation assay (aPCA)^{30,32}, an in-cell selection system. In aPCA, the protein domain

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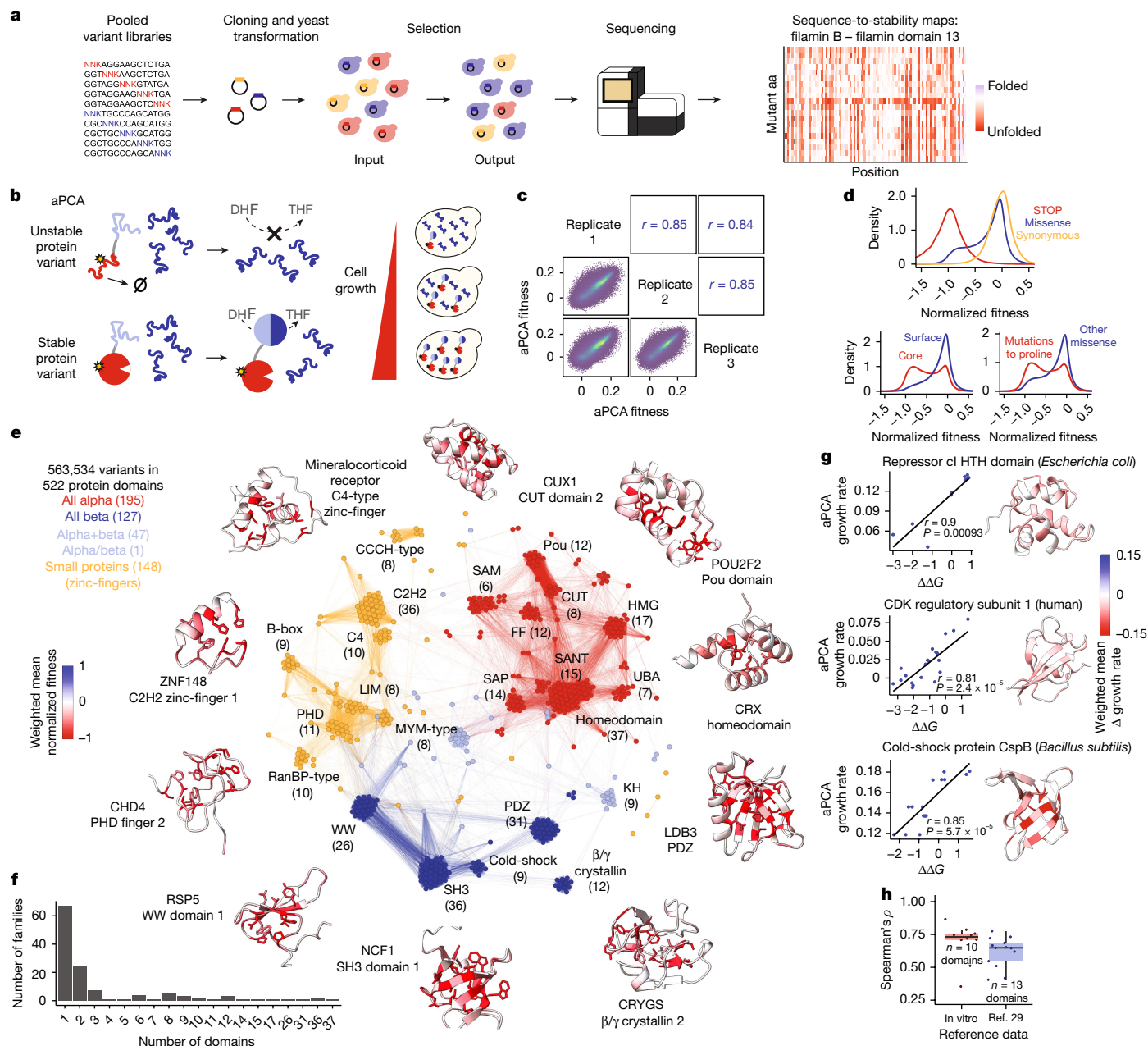


Fig. 1 | Mutating the human domainome. **a**, Experimental strategy for multiplexed generation of sequence-to-stability maps of human protein domains based on pooled cloning, transformation and selection of saturation mutagenesis libraries. **b**, Dihydrofolate reductase (DHFR) complementation assay to measure in vivo abundance of variant human protein domains (aPCA³⁰). Variants that cause unfolding and degradation of the target domain result in a decrease in concentration in the cell, leading to impaired cell growth. DHF, dihydrofolate; THF, tetrahydrofolate. **c**, Replicate correlations of aPCA fitness scores. Pearson's r is shown. **d**, Comparison of aPCA fitness score distributions of synonymous, missense and nonsense variants (top), core (relative surface accessible solvent area (rSASA) <25%) and surface variants (bottom left) and mutations to proline and other missense variants (bottom right). **e**, Structural similarity network of

domains that are retained in the final dataset. Nodes represent protein domains, and edges represent Foldseek hit probabilities in pairwise structural alignments. Colours correspond to structural classes. **f**, Distribution of number of domains per protein family in the library. **g**, Correlations between in vitro $\Delta\Delta G$ values and aPCA scores in three representative domains. Pearson's r and its corresponding P value are shown. **h**, Distribution of correlations (Spearman's ρ) between aPCA scores and in vitro $\Delta\Delta G$ values^{33,34} or stabilities measured in a high-throughput proteolysis assay²⁹. Boxes indicate the median and the first (Q1) and third (Q3) quartiles, the bottom whisker indicates the lowest value no further than Q1 minus 1.5 times the interquartile range (IQR), and the top whisker indicates the largest value no further than Q3 plus 1.5 times IQR.

of interest is expressed as a fusion with a fragment of an essential enzyme, and the concentration of this enzyme linearly determines the cellular growth rate over at least three orders of magnitude³⁰ (Fig. 1b). The effects of variants on protein abundance are quantified using high-throughput sequencing to measure the change in variant frequencies between the input and output cell populations in selection experiments (Fig. 1a). This strategy thus enables pooled cloning,

transformation and selection of hundreds of thousands of variants in diverse proteins in a single experiment (Fig. 1a).

In total, we performed 27 transformation, selection and sequencing experiments (3 independent biological replicates of 9 sub-libraries). After filtering (Methods), the final dataset consists of cellular abundance measurements for 563,534 variants in 522 protein domains, of which 503 are from human proteins (Supplementary Table 2).

Abundance measurements were highly reproducible (median Pearson's correlation coefficient, $r = 0.85$ between replicates for all variants; Fig. 1c and Extended Data Fig. 1b). They also correlated well with independent in vitro measurements of protein fold stability^{33,34} (median Spearman's $\rho = 0.73$ with folding free-energy changes ($\Delta\Delta G$), $n = 10$ domains; Fig. 1g,h and Extended Data Fig. 1d,e). Moreover, they correlated well with high-throughput stability measurements from protease sensitivity assays²⁹ (median $\rho = 0.65$, $n = 13$ domains; Fig. 1h and Extended Data Fig. 1f).

The 522 domains are structurally diverse, comprising 195 in the all-alpha structural class, 127 all-beta domains, 48 mixed alpha and beta domains and 148 metal-binding zinc-finger domains (Fig. 1e). In total, they cover 127 different domain families, including 14 families with 10 or more homologous domains (Figs. 1f and 2), and 97 families with only 1 or 2 domains (Fig. 1f). Together they comprise 2.1% of all proteins, 1.2% of all domains and 2.0% of all unique domain families in the human proteome. Two-hundred and seventy five of the domains are encoded by human disease genes with 108 domains containing annotated pathogenic variants. Across the dataset, mutations in the buried cores of the domains are more detrimental than mutations on their surfaces (Fig. 1d), with mutations to polar amino acids having stronger destabilizing effects in cores and mutations to hydrophobic residues having stronger effects on surfaces (Extended Data Fig. 1g). Mutations to proline are the most detrimental overall (Fig. 1d), in both core and surface residues, with highly destabilizing effects in beta strands and helices and milder effects in coils (Extended Data Fig. 1g,h). A subset of deep mutational scans of domains belonging to the most abundant families is shown in Fig. 2, and the full mutagenesis dataset is available in Supplementary Fig. 1.

Evaluation of VEPs

Our dataset represents a nearly fivefold increase in stability measurements for mutations in human proteins (Fig. 3a) and so provides an opportunity to evaluate how well computational VEPs predict changes in stability. Several general VEPs provide reasonable prediction of abundance changes (Fig. 3b,c), including the protein language model ESM1v¹² (median $\rho = 0.48$) and the deep generative model EVE¹¹ (median $\rho = 0.48$). Amongst tested dedicated stability predictors, the graph neural network ThermoMPNN³⁵ performs best (median $\rho = 0.50$; Fig. 3b,c), and is the best predictor overall. Notably, all tested stability predictors perform poorly on small zinc-finger domains that require metal binding for stability (Extended Data Fig. 2a). After excluding zinc-fingers, ThermoMPNN is still the best performing method overall (median $\rho = 0.57$; Extended Data Fig. 2a), even when evaluating protein domains with no homology to the Megascake dataset²⁹ used for ThermoMPNN training (median $\rho = 0.57$; Extended Data Fig. 2b).

Contribution of stability to fitness

Fold stability is one of many biophysical properties that contribute to protein function and protein sequence conservation during evolution. In particular, many proteins bind other proteins, nucleic acids and small molecules or catalyse enzymatic reactions, and these functions often trade off with stability³⁶. The extent to which selection on protein sequences is driven by changes in stability rather than other biophysical properties is an important open question³⁷.

To address this, we compared experimentally quantified stability to evolutionary fitness quantified by ESM1v across more than 500,000 variants in more than 500 domains (Methods). We find that protein stability accounts for a median of 30% of the variance in protein fitness when considering all domains (Extended Data Fig. 2e). However, the contribution of stability to fitness varies across domain families, with stability making a larger contribution to the fitness of all-beta domains (40% of the variance) than to that of all-alpha (25%) and mixed

domains (25%) (Extended Data Fig. 2f). This is consistent with the lower structural tolerance to mutations of beta sheets compared with alpha helices³⁸, and suggests that stability is a more important determinant of the fitness landscapes of all-beta domains. The overall contribution of stability to protein fitness also varies in domains with different molecular functions. For example, stability has a larger contribution to fitness in SH3, WW, PHD finger and Pointed protein-protein interaction domains, and a lower contribution in DNA-binding HMG-box domains, nuclear hormone-type C4 zinc-fingers and homeodomains (Extended Data Fig. 2g).

Identification of functional sites

Mutations in binding interfaces, active sites and allosteric control sites typically have larger effects on function than can be accounted for by changes in protein stability^{29,32,37}. For example, quantifying the effects of mutations on protein binding and abundance enables binding interfaces and allosteric sites to be comprehensively mapped^{32,39}. We reasoned that a similar approach could be used to identify functional sites in hundreds of domains by combining our abundance measurements with evolutionary fitness quantified by ESM1v.

For individual domains, protein abundance is non-linearly related to evolutionary fitness predicted by ESM1v (Fig. 3d and Extended Data Fig. 3a). We used sigmoidal curves to model this relationship ($n = 426$ domains, ESM1v fitness range > 10 , wild-type aPCA fitness percentile < 30 ; Fig. 3d and Extended Data Fig. 3a). The residuals to these fits identify mutations with larger or smaller effects on evolutionary fitness than can be accounted for by changes in stability (Fig. 3d-f).

This analysis identifies a total of 102,231 mutations with larger effects on evolutionary fitness than can be accounted for by changes in stability (24% of the total; two-tailed z-test false discovery rate (FDR) < 0.1 , normalized aPCA residual > 0.3). These mutations are enriched in known functional sites annotated in the Conserved Domains Database (CDD) (odds ratio (OR) = 2.72, Fisher's exact test $P < 2.2 \times 10^{-16}$, $n = 3,104$ functional sites in 2,800 residues, Fig. 3g and Extended Data Fig. 3b). Defining evolutionary functional sites as residues with a weighted mean residual of less than 0.3 identifies a total of 5,231 sites in 426 domains. These sites are strongly enriched in CDD-annotated sites (OR = 4.50, Fisher's exact test $P < 2.2 \times 10^{-16}$; Fig. 3h) and identify many known DNA-, RNA- and protein-binding interfaces (Fig. 3i,j). However, these evolutionary functional sites also include 1,942 sites in 180 domains without existing CDD annotations, and 1,873 additional sites in domains with other CDD annotations (Supplementary Table 3).

Of note, evolutionary functional sites without known annotations are located in closer proximity to annotated functional sites (median side chain heavy atom distance (d) = 3.62 Å) than other residues (median d = 6.93 Å, $P < 2.2 \times 10^{-16}$). Many of these therefore act as 'second-shell' residues contacting DNA-, RNA- and protein-binding interfaces (Fig. 3f,j and Extended Data Fig. 3c), where sequence changes may indirectly impact binding via energetic interactions with interface residues^{32,39}.

Contribution of stability to pathogenicity

Human Domainome 1 contains 3,652 variants with clinical annotations. Of these, 621 are classified as pathogenic or likely pathogenic (hereafter referred to as pathogenic), 322 are classified as benign or likely benign (hereafter benign) and 2,709 are classified as variants of uncertain significance (VUS), with 114 domains containing at least one pathogenic variant. Pathogenic variants are unevenly distributed across domains, with 75% of pathogenic variants contained in 25% of domains, and 41 domains containing only a single pathogenic variant (Extended Data Fig. 4a).

In total, 380 out of 621 pathogenic variants (61%) cause a detectable domain destabilization (one-tailed z-test, normalized aPCA fitness < 0 , FDR < 0.1) and 303 out of 621 (48%) are strongly destabilizing (FDR < 0.1 ,

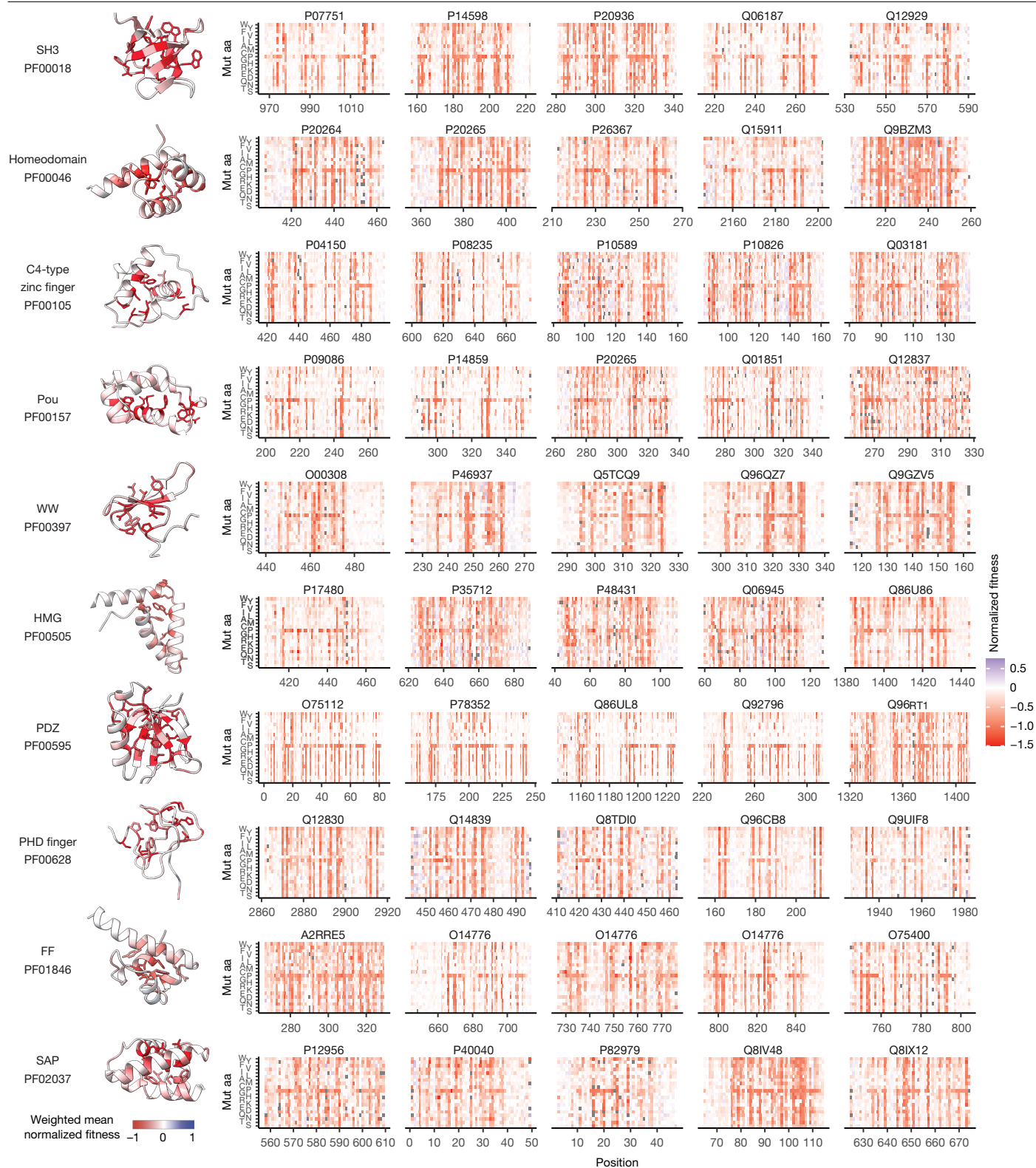


Fig. 2 | Deep mutational scans of protein homologues. Five examples of deep mutational scanning aPCA datasets of the most abundant protein families. Heatmaps depict the effects of mutating every residue in the protein domains (x axis) to all possible 19 amino acids (y axis). Mut aa, mutated amino acid.

normalized aPCA fitness < -0.3 ; Fig. 4a,b). This contrasts with 129 out of 322 (40%) and 50 out of 322 (16%) of benign variants, respectively. However, the association between pathogenicity and destabilization varies across domain families (Fig. 4a,b and Extended Data Fig. 4b). For example, many pathogenic mutations in β/γ crystallins that cause cataract disease are strongly destabilizing (13 out of 18, OR = 11.98 compared

with benign variants, Fisher's exact test $P = 1.99 \times 10^{-6}$; Fig. 4b). By contrast, a smaller proportion of pathogenic variants are strongly destabilizing in homeodomains (OR = 3.94, $P = 1.45 \times 10^{-7}$), HMG-box domains (OR = 2.10, $P = 0.045$) and CUT domains (OR = 1.65, $P = 0.55$) (Fig. 4b), all of which bind DNA, suggesting that many pathogenic variants in these domain families affect biophysical properties beyond stability, such as

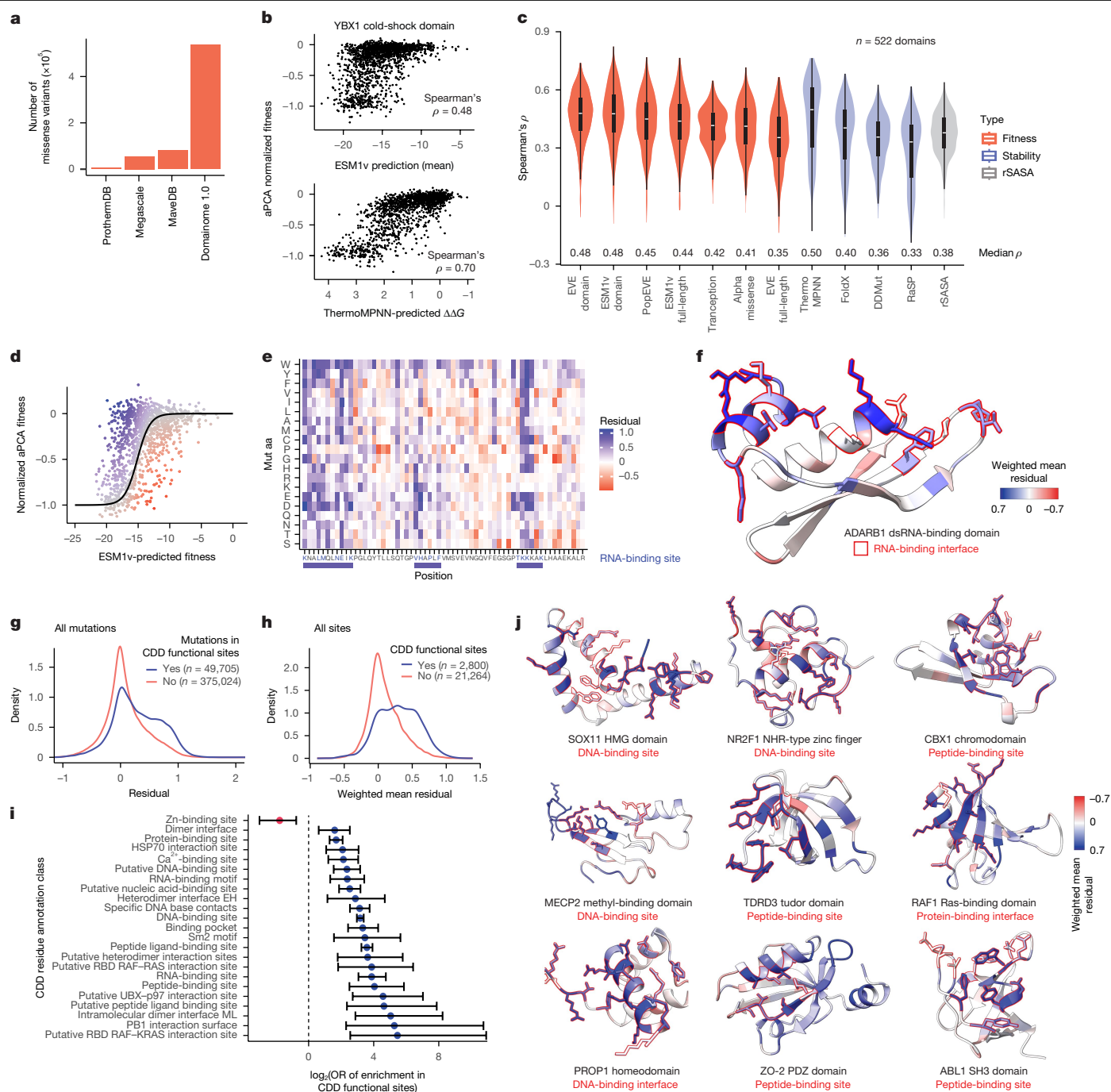


Fig. 3 | The contribution of protein stability to evolutionary fitness. a, Total number of stability measurements for human single missense variants in Human Domainome 1 compared with previous datasets of protein stability^{29,33,66}. **b**, Correlations between aPCA fitness and ESM1v predictions (top) or ThermoMPNN stability predictions (bottom) for the YBX1 cold-shock domain. **c**, Performance comparison of variant effect and protein stability predictors ($n = 522$ domains). The distribution of Spearman's ρ across all domains is shown, with the median ρ indicated above the x axis. Boxes indicate the median, Q1 and Q3, the bottom whisker indicates the lowest value no further than Q1 minus 1.5 times IQR, and the top whisker indicates the largest value no further than Q3 plus 1.5 times IQR. **d**, Sigmoidal curves to model the relationship between stability and evolutionary fitness of variants in the double-stranded RNA (dsRNA)-binding domain of ADAR1. Data points are coloured by their residuals to the fit. **e**, Heat map depicting the residuals to the fit of all measured variants. dsRNA-binding site

residues are marked with blue letters. Blue boxes capture all contiguous stretches of sequence containing the binding site. **f**, AlphaFold2-predicted structure of the dsRNA-binding domain of ADAR1, with residues coloured by the weighted mean residuals to the fit. Residues forming the dsRNA-binding site are marked with a red silhouette. **g**, Distribution of residuals for mutations in annotated functional sites, and in other protein residues. **h**, Distribution of weighted mean residuals of protein residues in annotated functional sites, and of other residues. **i**, Enrichment of evolutionary functional sites (residues with weighted mean residuals > 0.3) in several types of CDD-annotated functional sites. Error bars depict the 95% confidence interval for the OR of enrichment (Fisher's exact test). **j**, AlphaFold2-predicted structures of representative domains containing functional sites, with residues coloured by the weighted mean residuals. Residues belonging to CDD functional sites are marked with a red silhouette.

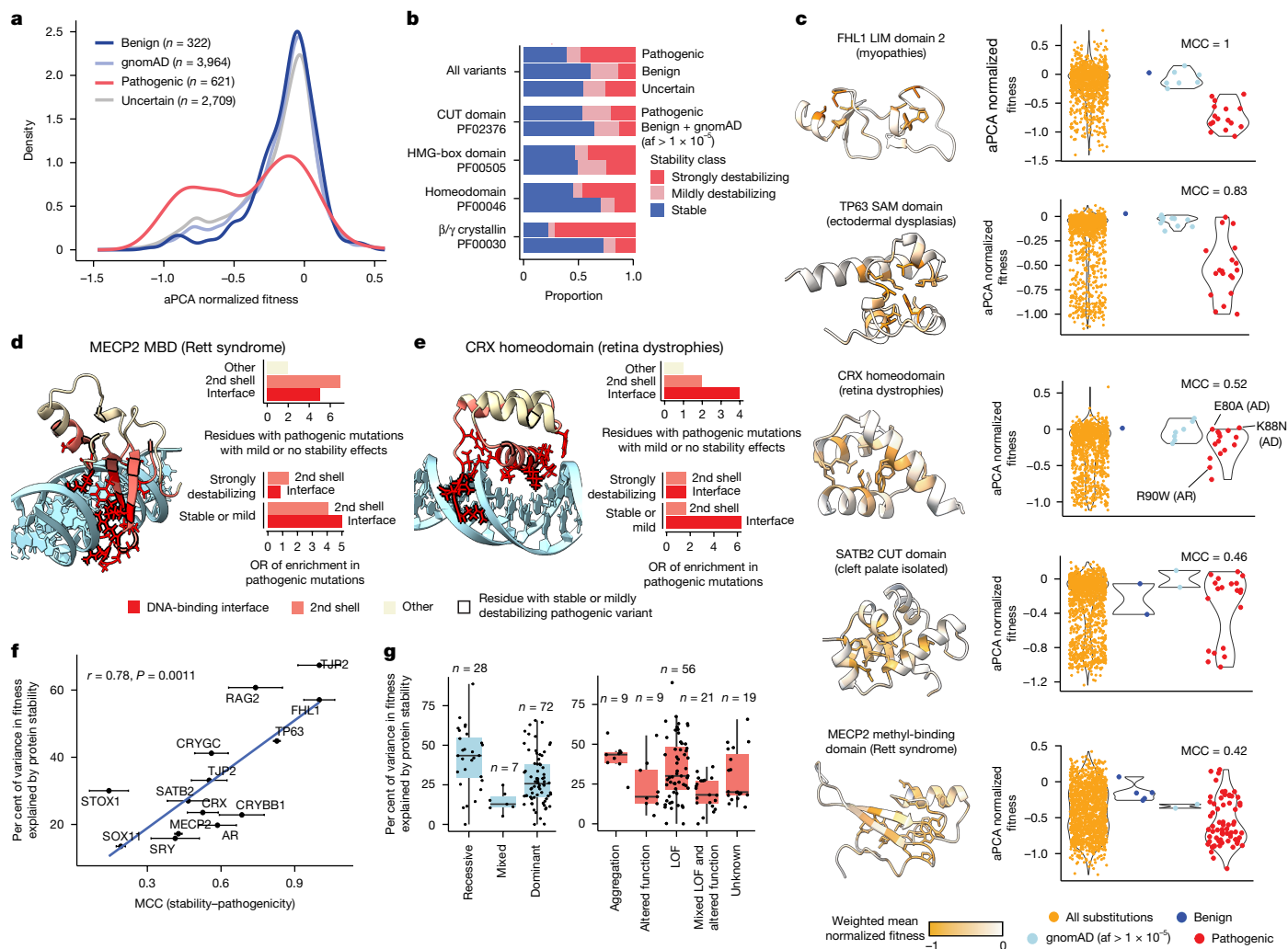


Fig. 4 | The contribution of protein destabilization to genetic disease.

a, Distributions of normalized aPCA fitness values of pathogenic, benign, uncertain and gnomAD variants (allele frequency $> 10^{-5}$). **b**, Proportions of stability classes (stable, mild or strongly destabilizing) in the full dataset (top bars) and in several protein families. **c**, Distributions of mutational effects on stability in representative examples of human disease protein domains, showing all measured variants (yellow), benign variants (dark blue), gnomAD variants with allele frequency (af) $> 1 \times 10^{-5}$ (light blue) and pathogenic variants (red). Specific CRX variants with their associated mode of inheritance are labelled. **d,e**, Spatial distribution of pathogenic mutations that do not strongly destabilize the MBD domain of MECP2 (**d**) and the CRX homeodomain (**e**). **f**, Correlation between the classification performance of stability on clinical variants

(measured using MCC), and the fraction of protein fitness explained by stability changes, for domains with at least 20 clinical and gnomAD variants (allele frequency $> 10^{-5}$). Error bars represent the s.d. of the MCCs calculated from resampled aPCA variant data ($n = 10$ times; Methods). Pearson's r and the corresponding P value are shown. **g**, Fraction of variance in protein fitness explained by stability changes in domains grouped by the mode of inheritance of their associated disorders (left) or their disease mechanisms (right). n refers to the number of domains. Boxes indicate the median, Q1 and Q3, the bottom whisker indicates the lowest value no further than Q1 minus 1.5 times IQR, and the top whisker indicates the largest value no further than Q3 plus 1.5 times IQR. LOF, loss of function.

DNA binding. Consistent with this variable association of destabilization with pathogenicity, aPCA has an overall lower performance than general VEPs in the classification of clinical variants across all domains, as the readout is specific for protein stability (Extended Data Fig. 4c,d).

We next quantified the relationship between stability and pathogenicity in all individual domains with at least 20 annotated clinical variants ($n = 17$). Stability is the major contributor to pathogenicity in some domains. In the LIM domain 2 of FHL1, stability is an excellent classifier of pathogenic variants that cause reducing body myopathy (Matthew's correlation coefficient (MCC) = 1; Fig. 4c), a disease caused by the accumulation of FHL1 aggregates⁴⁰. Similarly, the dominant ankyloblepharon–ectodermal defects–clefting syndrome is caused by mutations in the SAM domain of TP63 that lead to TP63 aggregation⁴¹. Accordingly, most pathogenic variants in the SAM domain of TP63 are destabilizing (MCC = 0.83; Fig. 4c).

By contrast, stability changes are a poorer predictor of pathogenic variants in other domains. A large proportion of mutations in the methyl-binding domain (MBD) of MECP2 that cause the dominant Rett syndrome are not destabilizing (MCC = 0.4; Fig. 4c). This suggests that many of these haploinsufficient variants interfere with the methylated-DNA-binding function of MECP2^{42–44} without affecting the overall stability of the domain. Indeed, pathogenic mutations in MECP2 MBD that are not strongly destabilizing are concentrated in its DNA-binding interface (OR = 5.00, $P = 0.09$; Fig. 4d), in second-shell (OR = 4.09, $P = 0.15$) and in positively charged surface residues (OR = 4.35, $P = 0.019$), probably leading to a loss of binding affinity. Similarly, multiple mutations in the CRX homeodomain that cause inherited retinal dystrophies are not strongly destabilizing (MCC = 0.52; Fig. 4c). These variants are also enriched in its DNA-binding interface (OR = 6.33, $P = 0.14$; Fig. 4d) and in positively charged sites (OR = 3.58,

$P = 0.11$). Notably, the mode of inheritance of mutations in CRX⁴⁵ correlates with their stability effects: whereas the recessive R90W is strongly destabilizing (Fig. 4c), the dominant K88N and E80A are stable (Fig. 4c), consistent with their described gain-of-function mechanisms⁴⁵. This again suggests that destabilization is the major disease mechanism for some proteins and diseases but is much less important for others.

Stability in dominant and recessive disorders

Comparing across all domains with at least 20 clinical variants, there is a substantial correlation between how well stability explains pathogenicity and how well stability explains evolutionary fitness, as quantified by ESM1v (Pearson's $r = 0.78$; Fig. 4f). We therefore used the correlation between stability and evolutionary fitness to rank all 114 domains with at least one known pathogenic variant (Extended Data Fig. 4f). Domains for which stability is highly predictive of fitness include many PHD finger and PDZ domains (Extended Data Fig. 4f). In these domains, we expect stability changes to be the major driver of pathogenicity. By contrast, stability only poorly predicts the fitness of homeodomains, HMG-box domains and nuclear hormone receptor-type zinc-finger domains (Extended Data Fig. 4f), suggesting that other molecular mechanisms will more frequently cause pathogenicity.

The contribution of stability to protein fitness also varies among genes with different modes of inheritance and disease mechanisms (Fig. 4g). Recessive diseases are strongly associated with loss of function, whereas dominant diseases can also be caused by additional mechanisms such as gain-of-function and dominant-negative effects or toxic aggregation^{46,47}. A median 44% of the variance in protein fitness is accounted for by stability changes in proteins mutated in recessive diseases, in contrast to only 26% in dominant disorders ($P = 1.1 \times 10^{-3}$, Wilcoxon rank sum test), suggesting that protein variants more frequently affect biophysical properties other than stability in dominant disorders. Indeed, loss-of-function and aggregation diseases are better explained by protein destabilization than 'altered-function' diseases associated with gain-of-function or dominant-negative mechanisms (Fig. 4g). Despite this overall association, however, we also find that within loss-of-function diseases, the extent to which destabilization explains protein fitness is variable (Fig. 4g). In summary, mutagenesis of more than 500 domains suggests that stability changes are an important cause of pathogenicity, but that this varies across proteins, with changes in stability being particularly important in recessive diseases.

Conservation of mutational effects

An important goal of establishing Human Domainome 1 is to quantify the extent to which mutational effects are conserved in structurally homologous proteins. Mutations in homologous sites might be expected to have very similar effects. However, mutations can also interact energetically, resulting in changes in mutational effects that depend on the sequence context, a phenomenon known as epistasis⁴⁸. If epistasis is prevalent, then mutational effects will be poorly conserved in divergent homologous proteins, limiting the extent to which experimental data from some proteins can be used to predict stability changes in homologues⁴⁹. Human Domainome 1 contains saturation mutagenesis data for 5 or more domains for 26 different domain families, enabling us to quantify the importance of epistasis for protein stability in structurally diverse protein folds.

To quantify the conservation of mutational effects, we fitted a thermodynamic model based on the Boltzmann partition function to all of the data for a domain family (Fig. 5a). The model assumes that mutations cause the same change in folding energy ($\Delta\Delta G$) in all homologous domains, and that the energetic effects of mutations combine additively with no specific epistasis (Fig. 5a,b). We first fitted this model to homeodomains, the most abundant domain family in the dataset, with 36 human homologues. The model provides very good prediction of

mutational effects across all 36 homeodomains (Pearson's $r = 0.78$ by tenfold cross-validation; Fig. 5c,d). The inferred free-energy changes can be considered homologue-averaged mutational effects, providing an energy model describing the entire family (Fig. 5e,f). A linear model provides similarly good performance but with biased prediction residuals ($r = 0.78$; Extended Data Fig. 5a,b). We additionally evaluated the performance of the Boltzmann energy model by leaving out single homeodomains from the training dataset ($n = 36$ models) and found similarly good performance (median Pearson's $r = 0.74$). Predictive performance was, as expected, better for domains with more similar sequence to the training dataset, but reasonable across a wide range of sequence divergence (Pearson's $r = -0.75$; Fig. 5g).

Energy models across domain families

Extending this analysis to all 26 families with at least 5 homologues in Human Domainome 1 (Fig. 5h) results in similarly accurate Boltzmann energy models for all families (median Pearson's $r = 0.80$, median percentage of explainable variance = 80.7%; Fig. 5i and Extended Data Fig. 6a,b). The performance on individual left-out homologues is also very good (median Pearson's $r = 0.66$, median percentage of explainable variance = 73.5%, Fig. 5j). For most domains, predictions were best for the domains most similar to the training dataset but were also good for domains with higher sequence divergence (Extended Data Fig. 7). Predictions using these energy models were better than those made with ThermoMPNN, the top-performing stability predictor on our dataset (Fig. 5j), and also had a good performance on stability deep mutagenesis scans generated using in vitro proteolysis selections²⁹ ($n = 38$ homologues, median Pearson's $r = 0.65$; Fig. 5k).

The excellent performance of these additive energy models is both useful and important: it demonstrates that epistasis makes only a small contribution to protein stability across these levels of sequence divergence. Combinatorial mutagenesis of individual proteins suggests a similar conclusion⁵⁰. The decay of predictive performance with sequence divergence does, however, suggest a role for epistasis in the evolution of protein stability. Indeed, we identify 25,410 mutations with evidence of epistasis as variants with large residuals to the energy model fits (FDR < 0.1 two-tailed z-test, $|\text{residual}| > 0.05 \text{ h}^{-1}$; Extended Data Fig. 8a,b). These epistatic variants are enriched in the buried cores of protein domains (OR = 2.71, Fisher's exact test $P = 1.05 \times 10^{-5}$) and depleted from protein surfaces (OR = 0.19, Fisher's exact test $P = 9.79 \times 10^{-14}$; Extended Data Fig. 8c) across the full range of measured stabilities (Extended Data Fig. 8d) and protein families (Extended Data Fig. 8e). This data suggests that genetic interactions are more important for the evolution of protein cores, consistent with these residues having a larger number of structural contacts than solvent-exposed residues (Extended Data Fig. 8f).

Prediction of homologues proteome-wide

Finally, the good performance of the Boltzmann energy models across all 26 families suggests that we can use them to provide proteome-wide stability predictions for these domain families. Using the models we made predictions for an additional 4,107,436 variants in 7,271 domains, including an additional 13,878 clinical variants, of which 1,310 are pathogenic, 951 are benign and 11,617 are variants of uncertain significance (Extended Data Fig. 9a–c). Consistent with the results for clinical variants with experimentally measured stability changes, 686 (52%) of these pathogenic variants are predicted to reduce protein stability (one-tailed z-test, scaled $\Delta\Delta G > 0$, FDR < 0.1, OR = 2.95 compared with benign), and 452 (34%) are predicted to have strongly destabilizing effects (scaled $\Delta\Delta G > 0.3$, FDR < 0.1, OR = 5.95 compared with benign; Extended Data Fig. 9d,e). Similarly to the fitness-level data, our energy models outperform stability predictors in pathogenicity prediction, and have a lower performance than general VEPs (Extended Data Fig. 9f). These

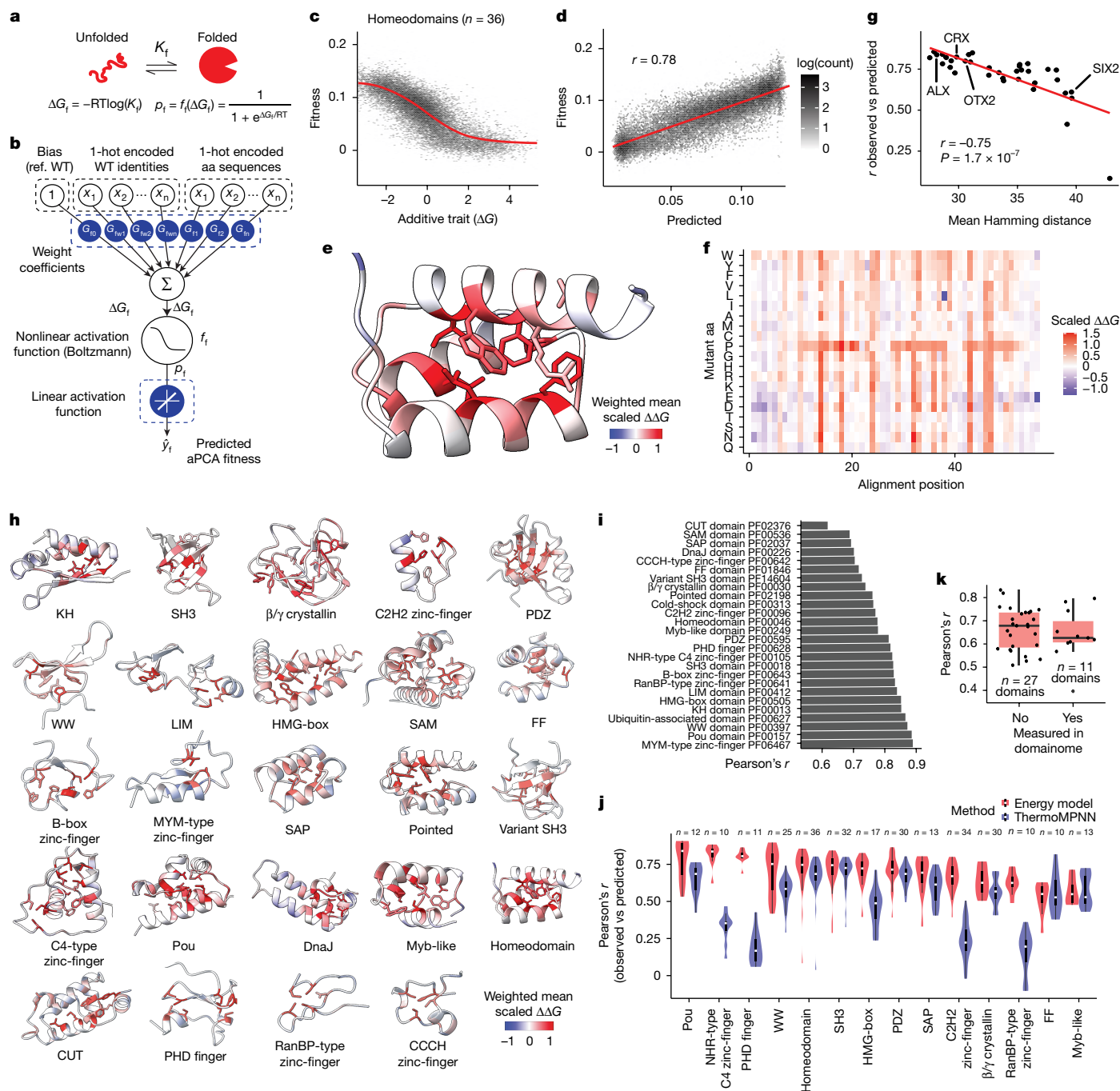


Fig. 5 | The genetic architecture of stability across protein families.

a, Two-state folding equilibrium and corresponding thermodynamic model. ΔG_f , Gibbs free energy of folding; K_f , folding equilibrium constant; p_i , fraction folded; f_i , nonlinear function of ΔG_i ; R , gas constant. **b**, Model architecture used to fit thermodynamic models to protein families. **c**, Relationship between predicted fitness and additive trait (ΔG) in a Boltzmann model fit to the homeodomain family (PF00046). **d**, Correlation between observed fitness values and MoCHI fitness predictions of the homeodomain family for test set variants (tenfold cross-validation; variants held out in any of the ten training folds are shown). **e**, Structure of a representative homeodomain, with residues coloured by the weighted mean homologue-averaged $\Delta\Delta G$ of mutations at each position. **f**, Heat map depicting inferred homologue-averaged folding $\Delta\Delta G$ defining the stability of the homeodomain fold. **g**, Performance of the energy model on held-out homeodomains as a function of the average genetic distance to the training set. Pearson's r and

its corresponding P value are shown. **h**, Structures of representative domains of all modelled families with residues coloured by the weighted mean homologue-averaged $\Delta\Delta G$ of mutations at each position. **i**, Summary of energy model performance on held-out variants across protein families (tenfold cross-validation). **j**, Summary of energy model performance on held-out human homologous domains across protein families ($n > 9$ homologues), compared with the performance of the top stability predictor (ThermoMPNN). n refers to the number of domains in human proteins. **k**, Summary of energy model performance on deep mutational scans of homologous domains generated using an in vitro proteolysis assay for stability determination²⁹. Boxes in **j**, **k** indicate the median, Q1 and Q3, the bottom whisker indicates the lowest value no further than Q1 minus 1.5 times IQR, and the top whisker indicates the largest value no further than Q3 plus 1.5 times IQR.

high-quality stability predictions provide a resource for the mechanistic interpretation of clinical variants for entire protein families and suggest a strategy for expanding Human Domainome 1 proteome-wide by experimentally mutagenizing representative examples for all families.

Discussion

We present here a large-scale experimental analysis of mutational effects in human proteins. Human Domainome 1 demonstrates the feasibility of quantifying human protein variant effects at scale and provides high-confidence measurements of changes in protein abundance for 563,534 variants in 522 structurally diverse protein domains. The dataset increases the number of stability measurements for human variants about fivefold and serves as a large and standardized reference dataset for the interpretation of clinical variants and for benchmarking and training computational methods.

An important goal of this project was to evaluate the contribution of protein stability to genetic disease and sequence evolution. We found that approximately 60% of pathogenic missense variants reduce protein stability, but that this varies quite extensively across domains and diseases, with stability changes being particularly important in recessive disorders. Similarly, we found that the contribution of stability to evolutionary fitness varies across protein families, with a larger contribution in all-beta protein domains compared with other fold types. How this relates to protein evolvability and the importance of trade-offs between stability and other molecular functions will be interesting avenues for future research. A further goal of establishing Human Domainome 1 was to quantify the conservation of mutational effects in structurally homologous proteins. Fitting additive energy models to the data for domain families revealed that mutational effects on stability are largely conserved in homologous domains, with a small contribution from epistasis that increases with sequence divergence. This energetic additivity enables proteome-wide prediction of stability changes for entire protein families and suggests an efficient strategy to complete a first draft of the human domainome by mutagenizing representative examples for every protein family.

To maximize diversity and experimental efficiency, Human Domainome 1 focused on structurally diverse small protein domains. An important caveat of this approach is the use of domains isolated from their native sequence context, and the extent to which mutational effects differ in full-length proteins is an important question for future work. Advances in DNA synthesis⁵¹, assembly⁵² and mutagenesis^{53,54} should facilitate this. Moreover, many domains did not yield sufficient signal in aPCA, indicating low stability or solubility, and other domains had mutational effects that were incompatible with two-state folding. In future work, it will be important to understand the behaviour of these domains in the assay and to test methods to increase the signal-to-noise such as altering domain boundaries, expression as full-length proteins, mutagenesis of exposed hydrophobic residues and the use of different reporters or solubilization tags. In this study, we have reported the effects of Human Domainome 1 variants on cellular protein abundance. However, the same libraries can be re-used in future work to quantify variant effects on additional molecular traits, for example abundance in different cellular contexts^{8,55,56}, *in vitro* fold stability²⁹, protein–protein^{57,58} and protein–nucleic acid⁵⁹ interactions, protein localization⁶⁰, aggregation⁶¹ and allostery^{32,39,62}. It will also be important to include more complex genetic variants such as insertions and deletions⁶³ and to extend to proteome-wide coverage, including extracellular and transmembrane proteins⁶⁴.

Human Domainome 1 is an important step in the comprehensive experimental analysis of human protein variants. It forms part of an ongoing global effort to determine the consequences of every mutation in every human protein and to produce reference atlases for the mechanistic interpretation of clinical variants⁷. Beyond human genetics, it is also part of a broader effort to produce large, well-calibrated

datasets that quantify how changes in sequence alter the biophysical properties of proteins. We believe that such multimodal biophysical measurements for millions of proteins and variants will enable machine learning approaches to be effectively brought to bear on the generative functions of molecular biology, allowing accurate prediction and engineering from sequence⁶⁵.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-024-08370-4>.

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Methods

Media

LB: 10 g l⁻¹ Bacto-tryptone, 5 g l⁻¹ yeast extract, 10 g l⁻¹ NaCl; autoclaved 20 min at 120 °C. YPDA: 20 g l⁻¹ glucose, 20 g l⁻¹ peptone, 10 g l⁻¹ yeast extract, 40 mg l⁻¹ adenine sulfate; autoclaved 20 min at 120 °C. SORB: 1 M sorbitol, 100 mM lithium acetate, 10 mM Tris pH 8.0, 1 mM EDTA; filter-sterilized (0.2 mm nylon membrane, ThermoScientific). Plate mixture: 40% PEG3350, 100 mM lithium acetate, 10 mM Tris-HCl pH 8.0, 1 mM EDTA pH 8.0; filter-sterilized. Recovery medium: YPD (20 g l⁻¹ glucose, 20 g l⁻¹ peptone, 10 g l⁻¹ yeast extract) + 0.5 M sorbitol; filter-sterilized. SC-URA: 6.7 g l⁻¹ yeast nitrogen base without amino acid, 20 g l⁻¹ glucose, 0.77 g l⁻¹ complete supplement mixture drop-out without uracil; filter-sterilized. SC-URA/ADE: 6.7 g l⁻¹ yeast nitrogen base without amino acid, 20 g l⁻¹ glucose, 0.76 g l⁻¹ complete supplement mixture drop-out without uracil, adenine and methionine; filter-sterilized. MTX competition medium: SD-URA/ADE + 200 µg ml⁻¹ methotrexate (BioShop Canada), 2% DMSO. DNA extraction buffer: 2% Triton X-100, 1% SDS, 100 mM NaCl, 10 mM Tris-HCl pH 8, 1 mM EDTA pH 8.

Library design, synthesis and cloning

We sampled PFAM-annotated human protein domains (version 34.0) of intracellular human proteins (not defined as ‘extracellular’, ‘transmembrane’ or ‘secreted’ in UniProt), prioritizing proteins that had at least one annotated pathogenic variant in ClinVar³. We additionally included protein domains with in vitro stability measurements to benchmark the assay. We designed the library in two rounds: we first designed and selected two libraries (A1 and B3) containing single mutants of a total of 485 domains that had been previously tested to grow as wild types. To design a second set of libraries (C1 to C7), based on the results of A1 and B3, we excluded domains without a well-defined hydrophobic core (not having at least 10% of residues with rSASA <25%) and disordered domains defined as having an average AlphaFold2 pLDDT < 50, indicative of protein disorder. This second set contained 631 domains that had been previously tested to grow as wild type, and 132 additional domains not previously tested. The domain sequences were codon optimized with emboss backtransseq⁶⁷ using the *Saccharomyces cerevisiae* codon usage table, and excluding the GCT alanine codon to prevent the appearance of HindIII restriction sites. The domain sequences were fused to 5' gggctgctcctaagaatggctagc and 3' aagcttgccgggtggcgggtctg constant regions containing NheI and HindIII restriction sites for cloning.

Libraries were synthesized by GCATbio using mMPS technology³¹. The mMPS platform employs a streamlined four-module process—identification, sorting, parallel synthesis, and recycling—to enable an efficient and flexible workflow using traditional phosphoramidite chemistry for oligonucleotide synthesis. Silicon carbide wafers were used as the substrate, with each microchip uniquely engraved with a QR code, serving as both a chip identifier and a record of the predetermined sequence information. This allowed for precise sequence identification and assembly post-synthesis. Standard and degenerate bases (N/K, representing a mixture of A, T, C and G, or T and G, respectively) were employed for synthesis. The degenerate bases were pre-mixed in an optimized ratio to ensure equal representation of each nucleotide at the targeted mutation sites. Library A1, containing sequences shorter than 128 nt, was synthesized using single-stranded oligonucleotides directly cleaved from the microchip. By contrast, the remaining eight libraries (B3, C1 to C7), composed of longer sequences (sequence lengths ranging from 179 to 341 bases), were constructed via polymerase cycling assembly (PCA)-PCR. Initially, the sequences were divided into 378,896 sub-sequences, each ranging from 60 to 80 nt, for synthesis using the mMPS system. The full-length DNA fragments of each library were then assembled from these sub-fragments using PCA-PCR.

Libraries were cloned at a variant coverage of ~100× or greater by restriction digestion and ligation into pGJJ162 (Supplementary Table 6), an aPCA assay plasmid where the DHFR3 fragment is fused at the C

terminus of the target protein domains and the fusion is driven by the CYC promoter, and the DHFR1,2 fragment is expressed at high levels as is driven by the GPD promoter.

Large-scale transformation and competition assay

Variant libraries were transformed in triplicate with a coverage of 20× or greater (with a mean coverage across libraries ~170×). For each transformation, we grew a 1 l YPDA culture of late log phase *S. cerevisiae* BY4741 cells (OD₆₀₀ ~ 0.8–1), collected the cells by centrifugation (5 min, 4,000g), resuspended in 43 ml SORB medium and incubated for 30 min on a shaker at room temperature. Then, 875 µl of 10 mg ml⁻¹ previously boiled (5 min, 100 °C) single-stranded DNA was added to the cells, followed by 17.5 µg library plasmid DNA and 175 ml plate mixture. The mix was incubated for 30 min on a shaker at room temperature. 17.5 ml DMSO were then added, and the cells were split in 50 ml Falcon tubes for 20 min heat shock at 42 °C. Following incubation, cells were pooled and collected by centrifugation, the supernatant was discarded with a pump, and cells were resuspended in 250 ml recovery medium and incubated at 30 °C for 1 h. Cells were then centrifuged for 3 min at 3,000g and transferred into 1 l SC-URA. Ten microlitres of this culture were immediately plated onto SC-URA selective plates to monitor transformation efficiency. The rest of the culture was incubated for one or two overnights at 30 °C.

SC-URA cultures were used to inoculate a 1 l culture of SC-URA-ADE at an OD₆₀₀ = 0.2–0.4, which was grown overnight (input culture). Cells from this culture were inoculated in 1 l SC-URA/ADE + 200 µg ml⁻¹ MTX to select stably expressed protein domain variants. The remaining input cells grown SC-URA/ADE were collected and frozen for DNA extraction. MTX cultures were left to grow overnight to an OD₆₀₀ = 1.6–2.5, corresponding approximately to 5 generations, collected and frozen for DNA extraction (outputs).

DNA extraction, plasmid quantification and sequencing library preparation

Total DNA was extracted from yeast pellets equivalent to 50 ml of cells at OD₆₀₀ = 1.6 as described in our previous work^{32,39}. Plasmid concentrations in the resulting samples were quantified by against a standard curve of known concentrations by qPCR, using oGJJ152 and oGJJ153 as qPCR primers that amplify in the origin of replication of the aPCA assay plasmid.

To generate the sequencing libraries, we performed two rounds of PCR amplification. In the first round, we used primer pools (oTB595+ and oTB748+; Supplementary Table 7) flanking the inserts that introduce frame-shifting nucleotides between the Illumina adapters and the sequencing region of interest. To maintain variant representation, we carried out eight 100 µl PCR1 reactions per sample, each of which starting with 125 million plasmid molecules that we amplified for 8 cycles. The reactions were column-purified (QIAquick PCR purification kit, QIAGEN), and the purified product was amplified further using the standard i5 and i7 primers to add the remainder of the Illumina adapter sequences and the demultiplexing indices (dual indexing) unique to each sample. We carried out a total of 8 100 µl PCR2 reactions per sample, each starting with 20–40 ng of purified product, that was amplified for 8 more cycles. The resulting amplicons were run on a 2% agarose gel to quantify the samples before pooling them for joint purification, and to ensure the specificity of the amplification and check for any potential excess amplification problems. The final libraries were size selected by electrophoresis on a 2% agarose gel, and gel-purified (QIAEX II Gel Extraction Kit, QIAGEN). The amplicons were subjected to Illumina paired-end 2 × 150 sequencing on a NextSeq2000 instrument at the CRG Genomics facility.

Sequencing data processing and normalization

FASTQ files from paired-end sequencing of all aPCA selections were processed with DiMSum⁶⁸ v1.2.11 (<https://github.com/lehner-lab/>)

Article

DiMSum) using default settings with minor adjustments. The option “barcodeIdentityPath” was used to specify a variants file in order to recover designed variants only (NNK mutations in the protein domains present in each sublibrary), and starting and final culture optical densities and selection times were specified to infer absolute growth rates across libraries.

Data filtering and normalization

We computed several quality control metrics on a per-domain basis. First, the wild-type position in the fitness distribution, as the difference between the wild-type and the 95th percentile of the distribution, divided by the difference between the 95th and 5th percentile in the distribution (fitness 90% range). Second, the Pearson’s correlation coefficient between biological replicates for missense variants. Third, the correlation between fitness and solvent accessibility for all variants in each domain. And fourth, the correlation between fitness and hydrophobicity of all variants in each domain (measured as the first principal component of a comprehensive table of amino acid properties⁶⁹). All structural calculations of the domains analysed are based on AlphaFold2 predictions obtained from <https://alphafold.ebi.ac.uk/>.

Compared to folded domains (more than 10% core residues and pLDDT > 50), disordered and domains without a well-defined hydrophobic core (see ‘Library design’) behave in a very distinct fashion in aPCA, with the wild type located in the middle of the fitness distribution, hydrophobicity negatively correlated with protein abundance, and narrow dynamic ranges resulting from typically small effects of mutations (Extended Data Fig. 1c). We thus excluded these domains from further analysis in the main text. Folded domains showed larger dynamic ranges, with the wild-type among the fittest variants, a negative correlation between mutation sensitivity and solvent accessibility, and a lack of correlation to hydrophobicity (Extended Data Fig. 1c). For folded domains, the four quality metrics described above were highly correlated with each other, and we combined them using principal component analysis. The first principal component explained 64% of the variance, and was used as a single quality metric to rank all domains.

We retained all domains ranked 600 or higher, with a missense Pearson’s $r > 0.485$, and with more than 50% of variants measured (with at least 10 counts in at least 1 replicate), resulting in 538 domains. We removed from this set 16 additional domains that were not compatible with 2-state folding, either showing many large effect abundance-increasing mutations (O75364_PFO0046_64, O75956_PFO9806_73, P10242_PFO0249_89, P52952_PFO0046_140, Q13263_PFO0643_205, Q86TZ1_PFI3181_60, Q8IX03_PFO0397_1, Q8NDW8_PFI3181_799, Q9Y2H9_PFI7820_968, Q9Y6M9_PFO5347_15), narrow fitness ranges (P35637_PFO0641_421, EHEE-rd2-0005_1, HHH-rd2-0133_1), or clear correlations to hydrophobicity (E9PAV3_PFI9026_2040, HEEH-rd3-0223_1, Q5VTD9_PFO0096_193), resulting in a final set of 522 domains.

The distributions of quality metrics for retained and discarded folded domains, and for disordered domains and folded domains without a well-defined core are in Extended Data Fig. 1c. Of the 478 discarded folded domains, 224 had slow growth rates as wild type ($< 0.075 \text{ h}^{-1}$), 176 had an unfit wild type (wild-type position < 0.4), 107 were negatively correlated with hydrophobicity (Pearson’s $r < -0.25$), and 158 were not correlated with solvent accessibility (Pearson’s $r < 0$) (we note that many of the discarded domains meet more than one of these criteria). Discarded domains are shorter (median length = 49 amino acids (aa)) than retained domains (length = 58 aa, $p = 1.87 \times 10^{-14}$, Wilcoxon rank sum test), and are enriched in zinc-finger domains (57% of zinc-finger domains discarded in contrast to 43%, 41%, and 39% for all-alpha, all-beta, and alpha+beta, respectively, $P = 2.33 \times 10^{-6}$, Fisher’s exact test). Multimerization status only affected success rate marginally, as 49% of domains from proteins that form complexes⁷⁰ were discarded, compared to 53% of those that do not engage in protein–protein interactions ($P = 0.2$, Fisher’s exact test). Finally, domains with yeast orthologues⁷¹ were less likely to be

retained (45%) than domains without yeast orthologues (53%, $P = 0.08$, Fisher’s exact test).

We normalized the growth rates and growth rate errors within each protein domain by linearly scaling the data such that the wild type-normalized fitness equals zero, and the 2.5th percentile of the distribution of growth rates of all missense variants plus the wild type is equivalent to a normalized fitness of -1 . In all box plots shown in the manuscript, the central line represents the median, the upper and lower hinges correspond to the first and third quartiles, with the upper whisker extending from the hinge to the largest value no further than $1.5 \times \text{IQR}$, and the lower whisker extending from the hinge to the smallest value no further than $1.5 \times \text{IQR}$.

Structural similarity network representation

We generated a structural distance matrix based on Foldseek⁷² hit probabilities. We computed all pairwise alignments between the domains in the final retained set by first creating a Foldseek database ‘foldseek_domainome_db’ containing all domains, and then searching all domains against the database using `foldseek easy-search *.pdb foldseek_domainome_db foldseek_easy_allvsall tmp --format-output “query,target,aln tmscore,qtm score,ttmscore,prob” --exhaustive-search TRUE -e inf`.

To visualize the similarity network, we imported the domains as nodes and the Foldseek probabilities as edges into Gephi⁷³, and applied two layout algorithms: first the Fruchterman Reingold algorithm to equilibrium, which resulted in a clear separation of the different SCOP classes, followed by a Force Atlas 2 layout algorithm (preventing node overlap), which accentuated the separation between the different domain family clusters within each SCOP class.

Comparison to reference $\Delta\Delta G$ datasets

To compare aPCA measurements with in vitro $\Delta\Delta G$ values, we used domains containing at least 10 variants in more than a single residue measured in vitro and in aPCA, and with a range of at least 2 kcal mol^{-1} for the in vitro measurements, and of $0.075 \text{ h}^{-1} \log$ growth rate units for the aPCA measurements ($n = 10$ domains). To compare aPCA measurements with $\Delta\Delta G$ values derived from high-throughput proteolysis deep mutational scans²⁹, we used protein domains with an overlap (overlapping length/total alignment length) of at least 80%, and an aPCA measurement range of $0.075 \text{ h}^{-1} \log$ growth rate units ($n = 12$).

The total number of human missense variants measured in the Megascala dataset²⁹ was retrieved from the high-confidence dataset excluding double mutants and domain duplicates (from different Protein Data Bank (PDB) entries and genetic backgrounds of the same PDB entry). We also quantified the total number of human missense variants measured in all abundance (VAMP-seq, aPCA) datasets available (as of 7 July 2024) in MaveDB⁶⁶ total number of human missense variants in ProthermDB³³.

Comparison to VEPs

We generated ESM1v¹² predictions (<https://github.com/facebookresearch/esm>) using the domain sequences alone as input, and using sliding windows across the human proteome (size = 1,000 aa, step = 250 aa). For each domain, we used the predictions corresponding to the window in which the domain is most centred. We obtained precomputed AlphaMissense^{11–14} predictions from https://console.cloud.google.com/storage/browser/dm_alphamissense. We obtained EVE¹¹, popEVE⁷⁴ and Tranception¹³ precomputed scores from <https://pop.evemodel.org/>. ‘EVE domain’ scores computed on high-coverage alignments generated specifically for domainome domains were kindly shared by A. Kollasch and D. Marks. We used precomputed RaSP scores⁷⁵, and generated DDMut⁷⁶ and ThermoMPNN³⁵ predictions with the aid of AlphaFold2 structure predictions. FoldX⁷⁷ predictions were obtained from <https://ftp.ebi.ac.uk/pub/databases/ProtVar/predictions/stability/>. We used Spearman’s ρ to quantify the relationship between the predictions and aPCA fitness. Domains with homology to protein

domains in the Megascap dataset were defined using hmmer (<http://hmmer.org/>) hmmscan against PFAM, and predictors were evaluated on a homologue-free set to prevent leakage from ThermoMPNN training.

To estimate the fraction of the variance in mutational effects on evolutionary fitness that are attributable to protein stability, we calculated the correlation between ESM1v fitness predictions on full-length protein sequences and aPCA scores. The correlation coefficient was adjusted by the measurement error of the aPCA scores according to the Spearman disattenuation formula:

$$R_{xy} = \frac{r_{xy}}{\sqrt{r_{xx}}}$$

where R_{xy} is the disattenuated correlation coefficient, r_{xy} is the observed correlation coefficient, r_{xx} is the mean correlation coefficient between aPCA biological replicates. This procedure was applied to both linear (Pearson's) and rank (Spearman's) correlation coefficients. The two were highly correlated ($r = 0.94$) and we report the Pearson's r -based version for ease of interpretation.

Analysis of functional sites

We carried out this analysis for domains where (1) the wild type is above the 30th percentile in the fitness distribution; and (2) the range between the 5th and 95th percentile of the distribution of ESM1v-predicted fitness is greater than 10 ($n = 426$ domains). We used sigmoid curves to model the relationship between normalized aPCA fitness (stability) and predicted function (ESM1v) of all variants in each individual domain, with an upper bound of 0 (wild-type aPCA-normalized fitness) and a lower bound of -1:

$$f_{\text{aPCA}} = -1 + \frac{1}{1 + e^{-(f_{\text{ESM1v}} - x_{\text{mid}})/\text{scal}}}$$

Where f_{aPCA} is the aPCA-normalized fitness, f_{ESM1v} is the ESM1v-predicted fitness, x_{mid} is the midpoint of the sigmoid and scal is the steepness parameter. To prioritize fitting the low stability variants, we weighted the fit by the aPCA-normalized fitness:

$$w_i = (\max(f_{\text{aPCA}}) - \min(f_{\text{aPCA}}) - (f_{\text{aPCA},i} + 1))^2$$

We used a two-tailed z-test to identify mutations whose effects on fitness cannot be accounted for by stability effects. We calculated z-scores as the aPCA residuals to the fit divided by the aPCA error, derived P values based on the normal distribution, and performed multiple testing correction using Benjamini-Hochberg's FDR. We additionally calculated per-residue mean residuals weighted by the aPCA-normalized fitness error.

We obtained residue-level functional site annotations corresponding to the CDD⁷⁸ using the InterPro API. Second-shell residues were defined as those with a minimum heavy atom distance of 5 Å to functional site residues.

Analysis of the stability effects of pathogenic variants

We identified destabilizing variants using a one-tailed z-test. z-Scores were calculated as the normalized fitness divided by the normalized fitness error, a P value was derived on the basis of a normal distribution, and FDR multiple test correction was applied. Destabilizing variants were defined as variants with $\text{FDR} < 0.1$, and strongly destabilizing variants as $\text{FDR} < 0.1$ and a normalized aPCA fitness < -0.3 . We obtained clinical variant annotations from ClinVar (January 2024 version) and from the UniProt API. Variant annotations were highly consistent between the two sources, and were merged. We tested for enrichments of clinical variant classes in stability classes using a two-tailed Fisher's exact test.

To estimate to what extent destabilization explains pathogenic mutations in individual domains, we used MCC. To increase statistical power,

we included gnomAD variants with an allele frequency $>10^{-5}$ as benign. We estimated the errors in MCCs by resampling the dataset based on the mean and errors of aPCA scores, 10 times. To analyse the distribution of pathogenic mutations in the MECP2 methyl-binding domain and the CRX homeodomain, we generated AlphaFold3 predictions⁷⁹. The sequences of the domains and DNA strands used for the predictions are available as Supplementary Table 8. DNA-binding interface residues were defined using getcontacts⁸⁰, and second-shell residues were defined as those with a minimum heavy atom distance of 5 Å to binding interface residues.

To extend the analysis to a larger number of domains with small numbers of pathogenic variants, we used the fraction of variance in ESM1v-predicted fitness explained by aPCA scores as described above. MCCs based on clinical and gnomAD variants correlated well with the fraction of evolutionary fitness explained by stability effects in domains with at least 20 clinical and gnomAD variants ($r = 0.78$), validating the approach. We estimated errors in MCCs by resampling the aPCA fitness dataset using the mean and error of each variant, and recalculating the MCCs on clinical variants ($n = 10$ samples). To further validate the estimates of the contribution of stability to fitness, we estimated the fraction of variance in ESM1v-predicted fitness explained by thermoMPNN predicted stabilities for non-zinc-finger domains with at least 1 pathogenic variant ($n = 86$). These correlated well with the original estimates derived using our abundance data (Pearson's $r = 0.81$; Extended Data Fig. 2d).

Modes of inheritance and mechanisms of disease information were obtained from OMIM (<https://www.omim.org/>). To analyse the contribution of stability changes to disease according to mode of inheritance and mechanism of action controlling for protein composition (Extended Data Fig. 4e), we modelled the contribution of secondary structure and the percentage of core residues to the scores using linear models. We then extracted the model residuals as the composition-corrected scores.

Clinical variant classification performance comparisons

We used the pROC R package⁸¹ to generate receiver operating characteristic (ROC) curves and calculate ROC area under the curve. We incorporated gnomAD variants with an allele frequency $>10^{-5}$ as benign. To test the performance when combining structural and sequence features (secondary structure, rSASA, wild-type residue, mutated residue), ESM1v and aPCA scores, we used generalized linear models in R. In addition to training and evaluating on the full dataset, we trained the logistic regression models with 90% of the data and evaluated on the remaining 10% unseen data.

Thermodynamic modelling of protein domain families

We used MoCHI⁸² to fit 2-state thermodynamic models on a per-family basis (for families with at least 5 human homologues and variants with a mean count >29). We specified a neural network architecture consisting of a single additive trait layer for shared folding energies across the family and a shared linear transformation layer. We used a 'TwoState-FractionFolded' transformation derived from the Boltzmann distribution function that relates energies to proportions of folded protein. To map homologous positions across families, we used PFAM alignments of human domains. To input into MoCHI, we recoded each domain wild-type sequence as an indel sequence as long as the PFAM alignment, plus additional positions in the alignment with variants that encode the wild-type identity of each homologue. Mutations in each domain were encoded in the corresponding alignment position. This design allows MoCHI's one-hot encoding of both the mutations and the wild-type identities simultaneously, and the joint fitting of mutation $\Delta\Delta G$ and starting ΔG of each homologue. We trained the model using a tenfold cross-validation approach and evaluated the performance on held-out data.

We additionally fitted a linear model, and a 'TwoStateFractionFolded' model with shared energies across all homologues of a family but with

domain-specific linear transformations that account for potential differences in solubility of the folded and unfolded states between domains of the same family. These models resulted in highly correlated inferred energies to the original Boltzmann model (Pearson's $r = 0.972$ for the linear model, and $r = 0.938$ for the Boltzmann model with domain-specific linear transformations) and similar performances on held-out data across families. We discarded the linear model due to highly biased residuals, and chose the 'TwoStateFractionFolded' model without homologue-specific linear scaling as the simplest of the remaining models. To compare across protein families, energies were rescaled such that the 2.5th percentile of the distribution of energies is equivalent to a scaled $\Delta\Delta G = -1$ and the wild type to a $\Delta\Delta G = 0$.

We further evaluated the performance of the TwoStateFractionFolded model in families with at least ten homologues by training the model leaving out a single domain at a time. The genetic distance (hamming distance and BLOSUM62 distance) of each left-out homologue to each of the homologues that went into training was calculated and averaged, to compare to model performance. The fraction of explainable variance in aPCA scores accounted for by the models was estimated as the R^2 between observed and predicted fitness divided by the R^2 between replicates.

Epistasis analysis of protein domain families

We identified epistatic variants with large residuals to the MoCHI fits using a two-tailed z-test. We calculated z-scores as the residuals to the fit divided by the aPCA fitness errors, derived P values on the basis of a normal distribution, and applied FDR multiple testing correction. Epistatic mutations were defined as those with $|\text{residual}| > 0.05$ and $\text{FDR} < 0.1$. Enrichments of alignment sites in epistatic variants were calculated, and significantly enriched sites were identified using a two-tailed Fisher's exact test. Epistatic sites were defined as those with a $\log_2(\text{OR}) > 1.5$ and a Fisher's Exact test $\text{FDR} < 0.05$. We classified alignment sites as core residues (rSASA $< 25\%$ in at least 75% of homologues), surface residues (rSASA $> 25\%$ in at least 75% of homologues), or changing residues (the rest). We defined contacts using getcontacts⁸⁰.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All DNA sequencing data have been deposited in the Gene Expression Omnibus under the accession GSE265942. All aPCA measurements and their associated errors are available as Supplementary Table 2, including quality ranking by domain for data filtering. Weighted mean residuals of the comparisons between abundance and evolutionary fitness predictions are available as Supplementary Table 3. Homologue-averaged $\Delta\Delta G$ values and their associated errors mapped to homologous domains proteome-wide are available as Supplementary Table 4. aPCA scores and matching VEPs are available as Supplementary Table 5. Publicly available data sources for analysis: domain annotations and alignments were obtained from PFAM; AlphaFold2 structures were obtained from the AlphaFold Protein Structure Database; CDD functional site annotations were obtained from InterPro through the InterPro API; gnomAD v4 variants were downloaded from <https://gnomad.broadinstitute.org/>; and clinical variants were obtained from ClinVar and UniProt using the proteinsAPI. We used precomputed VEPs from AlphaMissense, EVE, popEVE, Tranception, FoldX and RaSP.

Code availability

Source code used to perform all analyses and to reproduce all figures in this work is available at <https://github.com/lehner-lab/domainome>. Files required to reproduce the analyses can be downloaded at <https://doi.org/10.5281/zenodo.11043642> (ref. 83).

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Author contributions A.B. designed the libraries and performed all experiments and analyses. X.J. and Y.S. synthesized the variant libraries. A.B. and B.L. conceived the project, designed analyses and wrote the manuscript, with input from all authors.

Competing interests B.L. is a founder and shareholder of ALLOX. The other authors declare no competing interests.

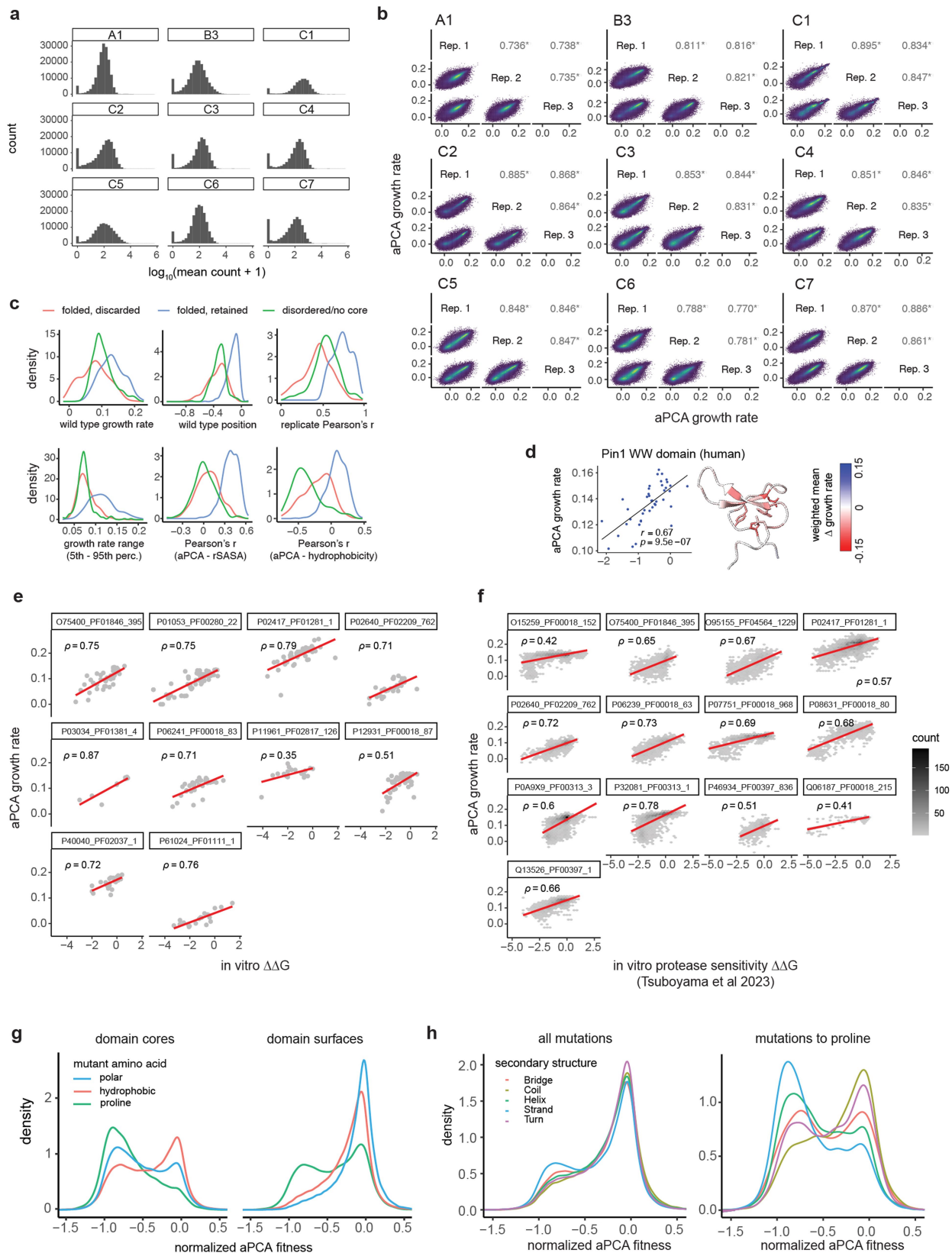
Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41586-024-08370-4>.

Correspondence and requests for materials should be addressed to Ben Lehner.

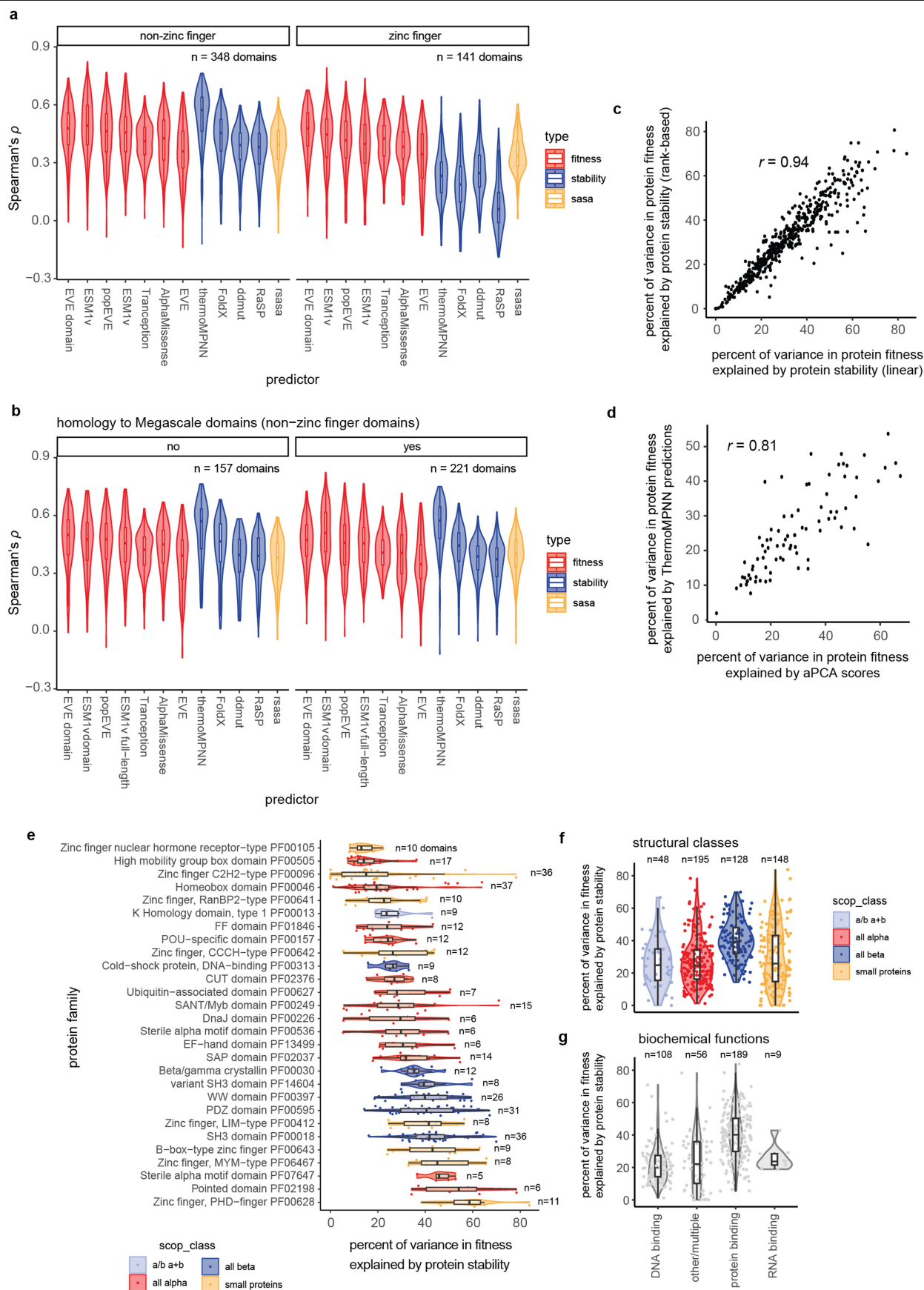
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Extended Data Fig. 1 | See next page for caption.

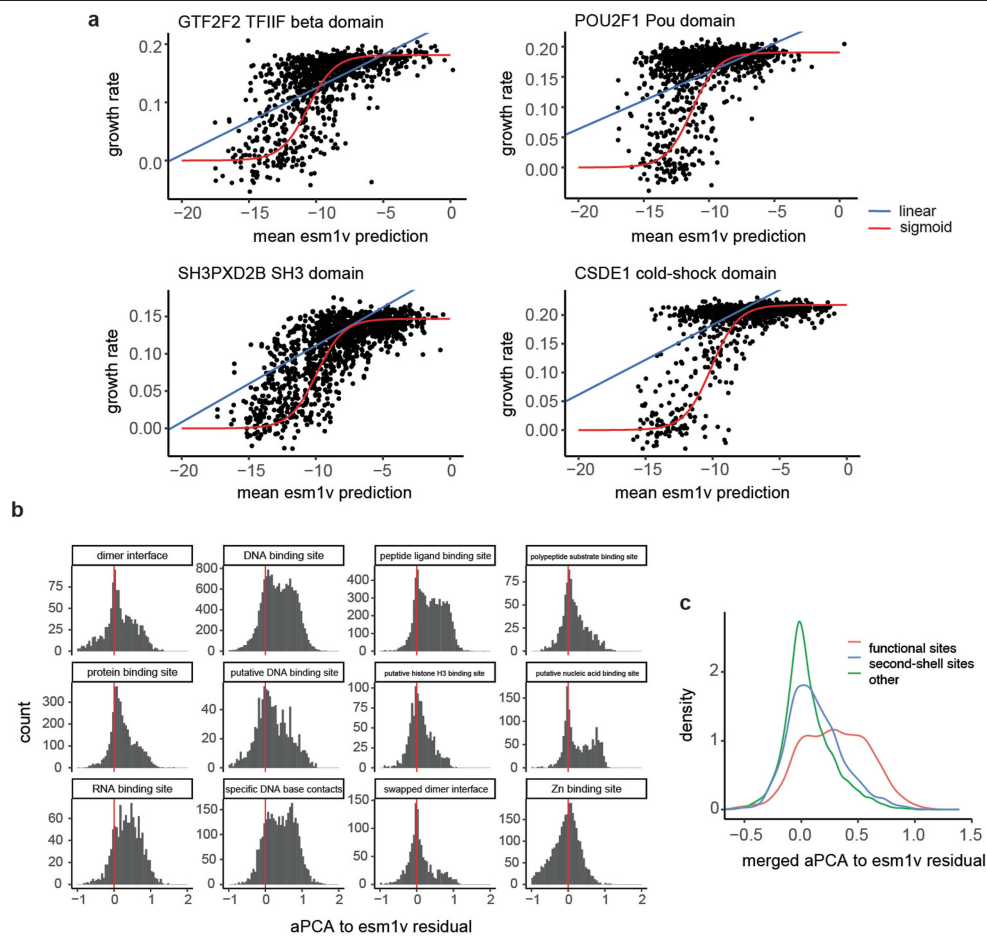
Extended Data Fig. 1 | Mutating the human domainome at scale. **a**, Histograms depicting sequencing coverage of the 9 site saturation mutagenesis libraries selected in this study. **b**, Correlations (Pearson's r) between replicates for the 9 libraries. **c**, Distributions of quality control metrics in folded domains that were retained for further analyses, in folded domains excluded from further analysis, and in domains with less than 10% core residues or disordered. **d**, Correlations between in vitro $\Delta\Delta G$ s and aPCA scores in the human Pin1 WW domain. **e**, Correlations (Spearman's ρ) between aPCA scores and in vitro $\Delta\Delta G$ s for individual protein domains. **f**, Correlations (Spearman's ρ) between aPCA scores and $\Delta\Delta G$ s derived through a high-throughput proteolysis assay for individual protein domains. **g**, Distributions of stability effects of mutations in protein cores and surfaces across the entire Domainome 1. Colours represent the physicochemical properties of the mutated amino acids (polar/charged = D, E, K, N, Q, R, S, T, hydrophobic = A, F, I, L, M, V, W, Y). **h**, Distributions of stability effects of mutations in different secondary structure types.



Extended Data Fig. 2 | See next page for caption.

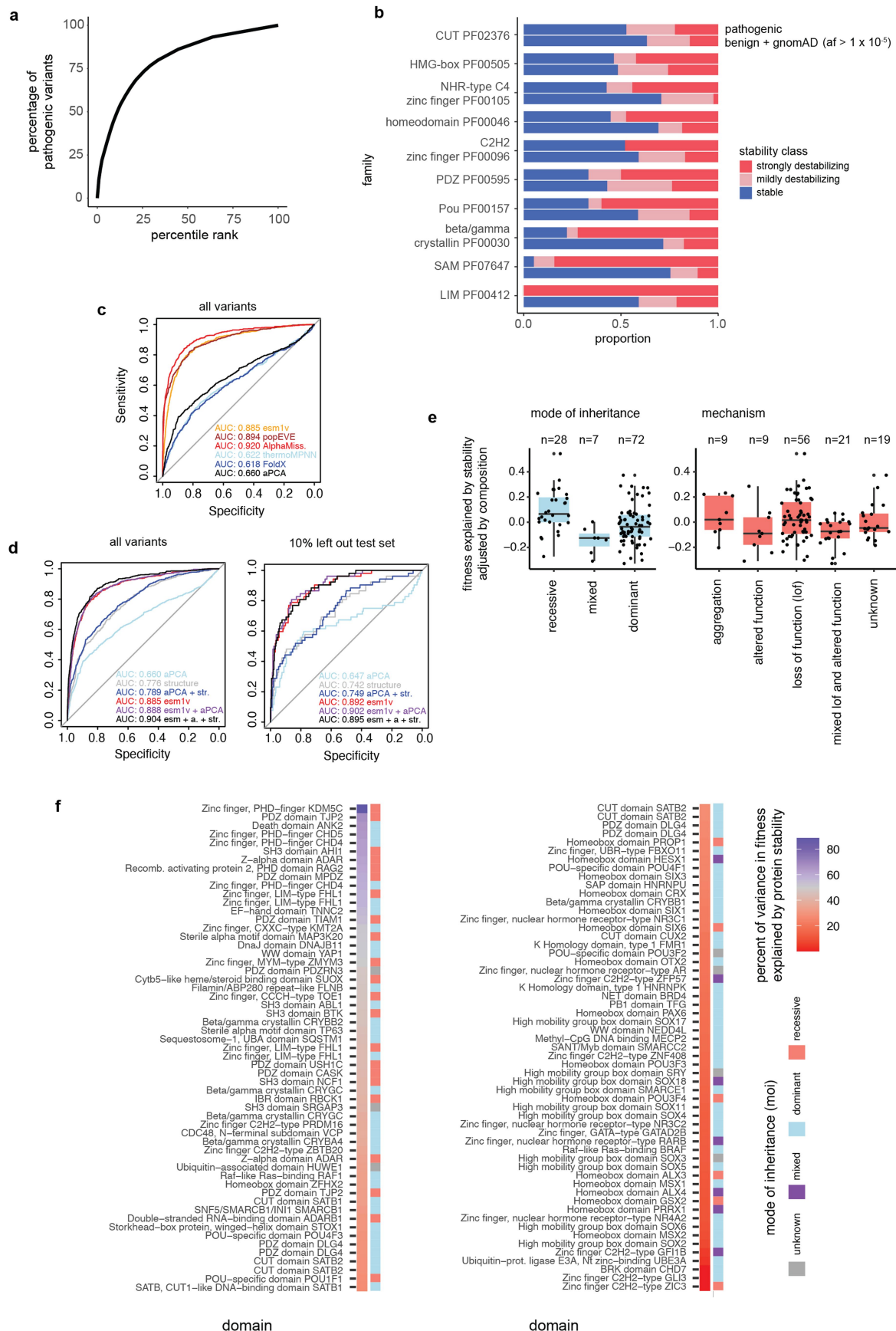
Extended Data Fig. 2 | The contribution of protein stability to evolutionary constraint. **a**, Comparison of performance of variant effect and protein stability predictors in non-zinc finger domains (left, $n = 348$ domains) and zinc finger domains (right, $n = 141$ domains). The distribution of Spearman's ρ across all domains is shown for each predictor. **b**, Comparison of performance of variant effect and protein stability predictors in non-zinc finger domains with (right, $n = 221$ domains) and without (left, $n = 157$ domains) homology to protein domains in the Megascale dataset used for ThermoMPNN training. **c**, Correlation between Pearson's r and Spearman's ρ -based variance explained estimates of the contribution of stability to protein evolutionary constraint. **d**, Correlation

between the contribution of stability to protein fitness estimated using aPCA data and using ThermoMPNN³⁵ predictions, for non-zinc finger domains containing at least one pathogenic variant. **e**, Quantification of the degree of sequence constraint attributable to protein stability across protein families with at least 5 measured homologues. Colours correspond to SCOP structural classes. **f, g**, Quantification of the degree of sequence constraint attributable to protein stability across domains grouped by their SCOP structural classes (**f**) and molecular functions (**g**). Boxes in **a**, **b**, **e**, **f**, and **g** indicate the median, Q1 and Q3; the lower whisker indicates the lowest value no further than $Q1 - 1.5 \times IQR$; and the upper whisker indicates the largest value no further than $Q3 + 1.5 \times IQR$.



Extended Data Fig. 3 | Identifying evolutionary functional sites using language models and abundance measurements. a, Representative examples of linear model and sigmoid model fits to the relationship between ESM1v predictions and abundance. **b,** Residuals to the fit of protein domain variants in

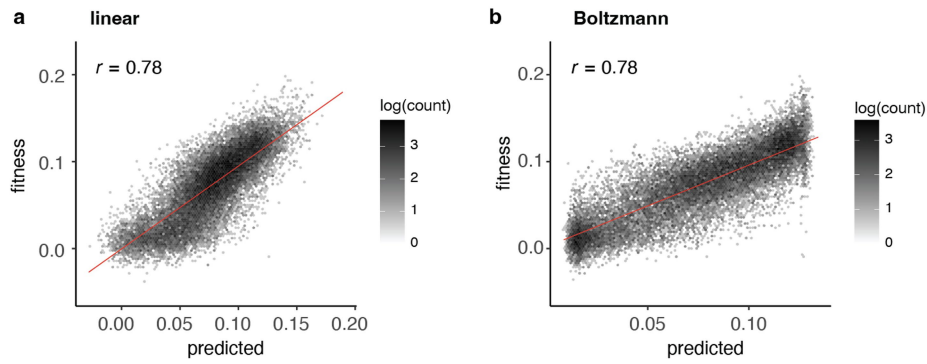
several classes of annotated functional sites. Classes of functional sites with at least 50 residues annotated across the full domainome dataset are shown. **c,** Distribution of abundance to ESM1v residuals in functional sites, second shell sites, and other sites, for domains with at least 1 annotated functional site in CDD.



Extended Data Fig. 4 | See next page for caption.

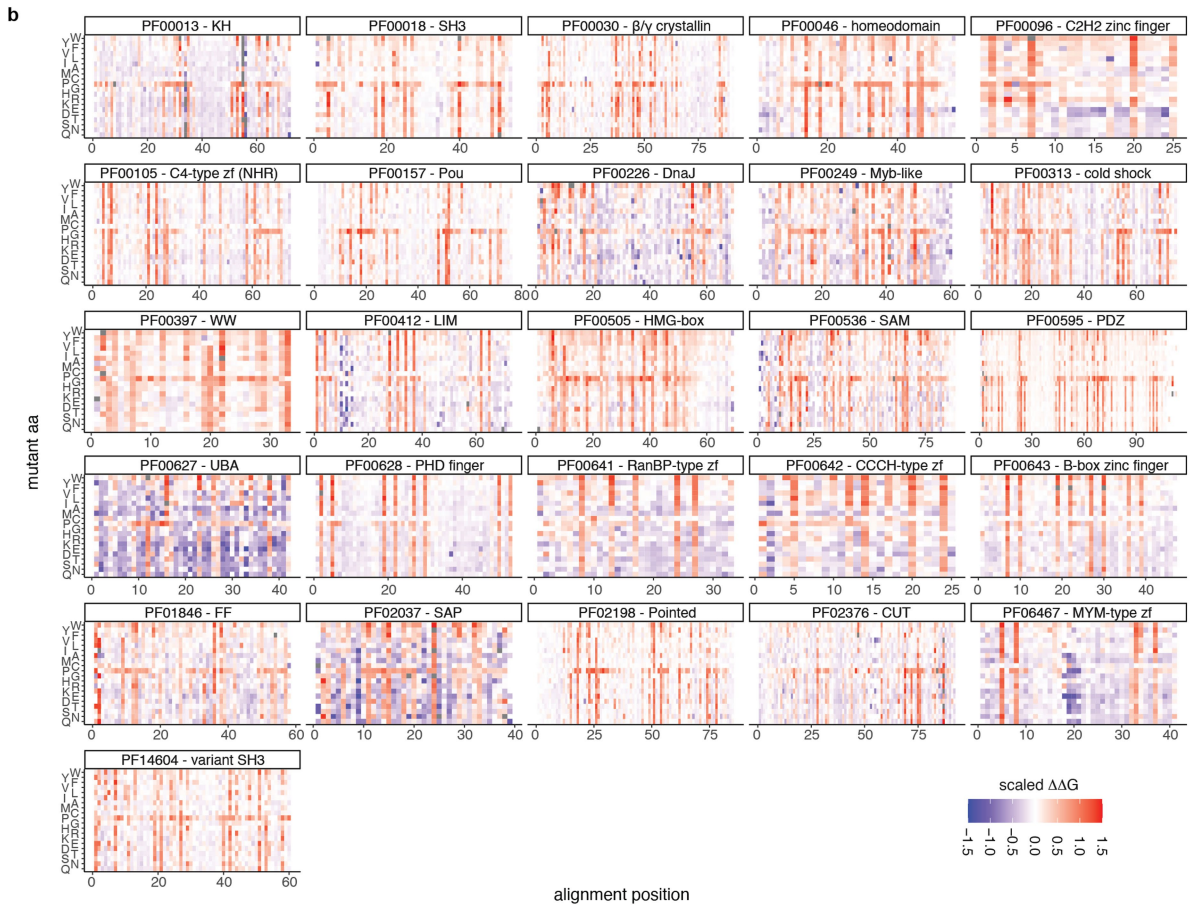
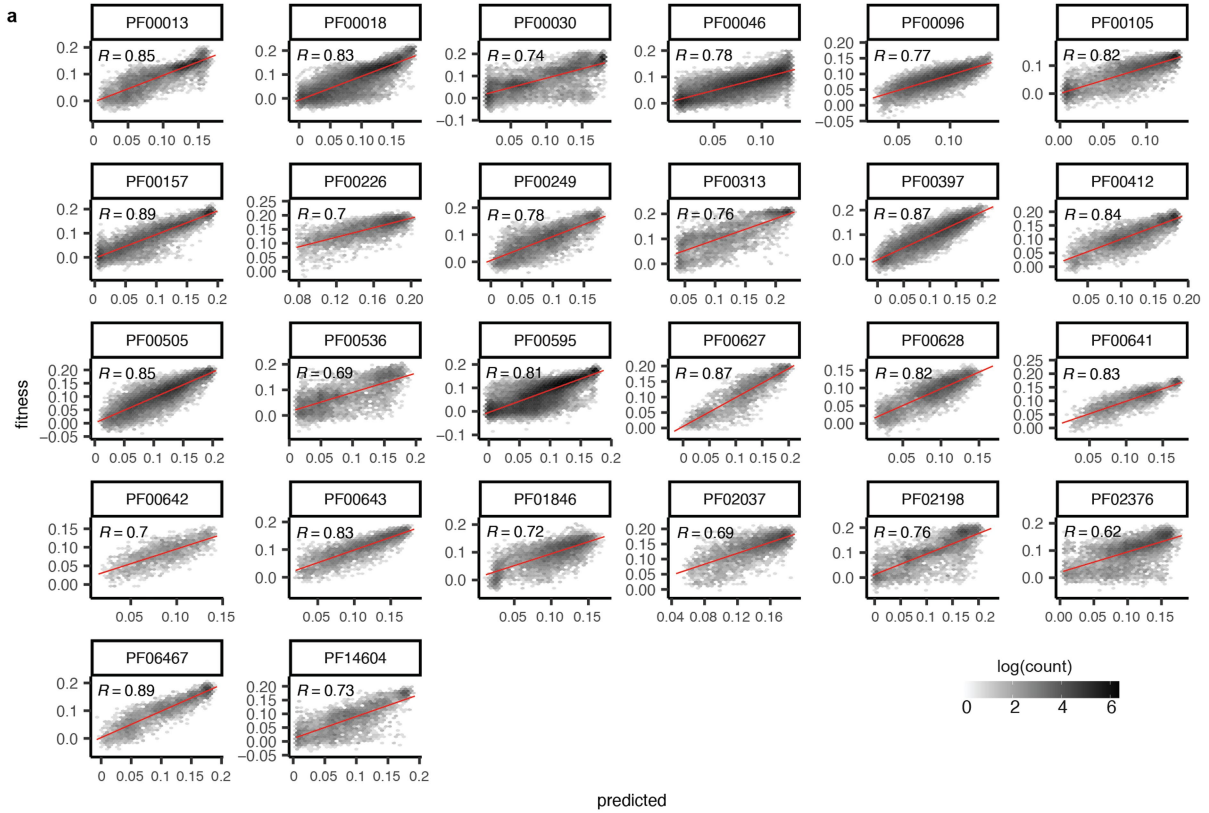
Extended Data Fig. 4 | The contribution of protein destabilization to genetic disease. **a**, Gini plot depicting the cumulative number of pathogenic variants present in Domainome 1 domains ranked by pathogenic variant counts (from highest to lowest). **b**, Proportions of non-destabilizing, mildly destabilizing ($\text{FDR} < 0.1$, $0 > \text{normalized aPCA fitness} > -0.3$), and strongly destabilizing ($\text{FDR} < 0.1$, $\text{normalized aPCA fitness} < -0.3$) variants in the pathogenic and benign sets, for several protein families. gnomAD variants with an allele frequency (af) $> 1 \times 10^{-5}$ were included as benign. Families with at least 10 variants in the pathogenic and in the benign classes are shown. **c**, ROC curves comparing clinical variant classification performance of aPCA data and variant effect or stability predictors. **d**, ROC curves comparing clinical variant classification performance of aPCA,

ESM1v, protein structural features (secondary structure, rSASA, wild-type aa, mutant aa), and combinations of these using logistic regression (see Methods). **e**, Distributions of the proportions of variance in fitness explained by stability changes corrected by protein composition differences across domains. Protein domains are split according to the mode of inheritance of pathogenic mutations (left) or their disease mechanisms (right). Boxes indicate the median, Q1 and Q3; the lower whisker indicates the lowest value no further than $Q1 - 1.5 \times \text{IQR}$; and the upper whisker indicates the largest value no further than $Q3 + 1.5 \times \text{IQR}$. **f**, Ranking of all 114 domains containing at least 1 pathogenic variant by the percentage of variance in fitness explained by stability changes (top left domains = highly explained by stability, bottom right domains = poorly explained by stability).



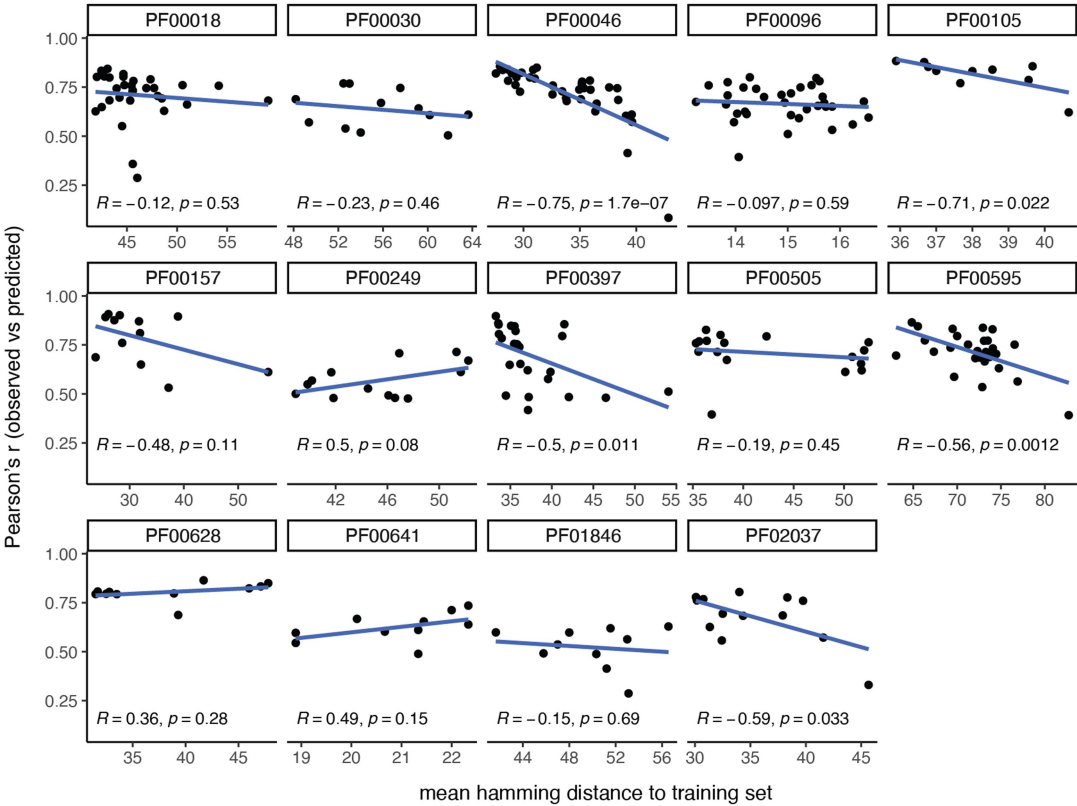
Extended Data Fig. 5 | Comparison of linear and Boltzmann energy model fits to a protein family. Correlations between observed fitness values and energy model predictions for test set variants held out in any of the 10 training

folds, for an energy model with a linear activation function (**a**) or a Boltzmann activation function (**b**) at the additive trait layer.

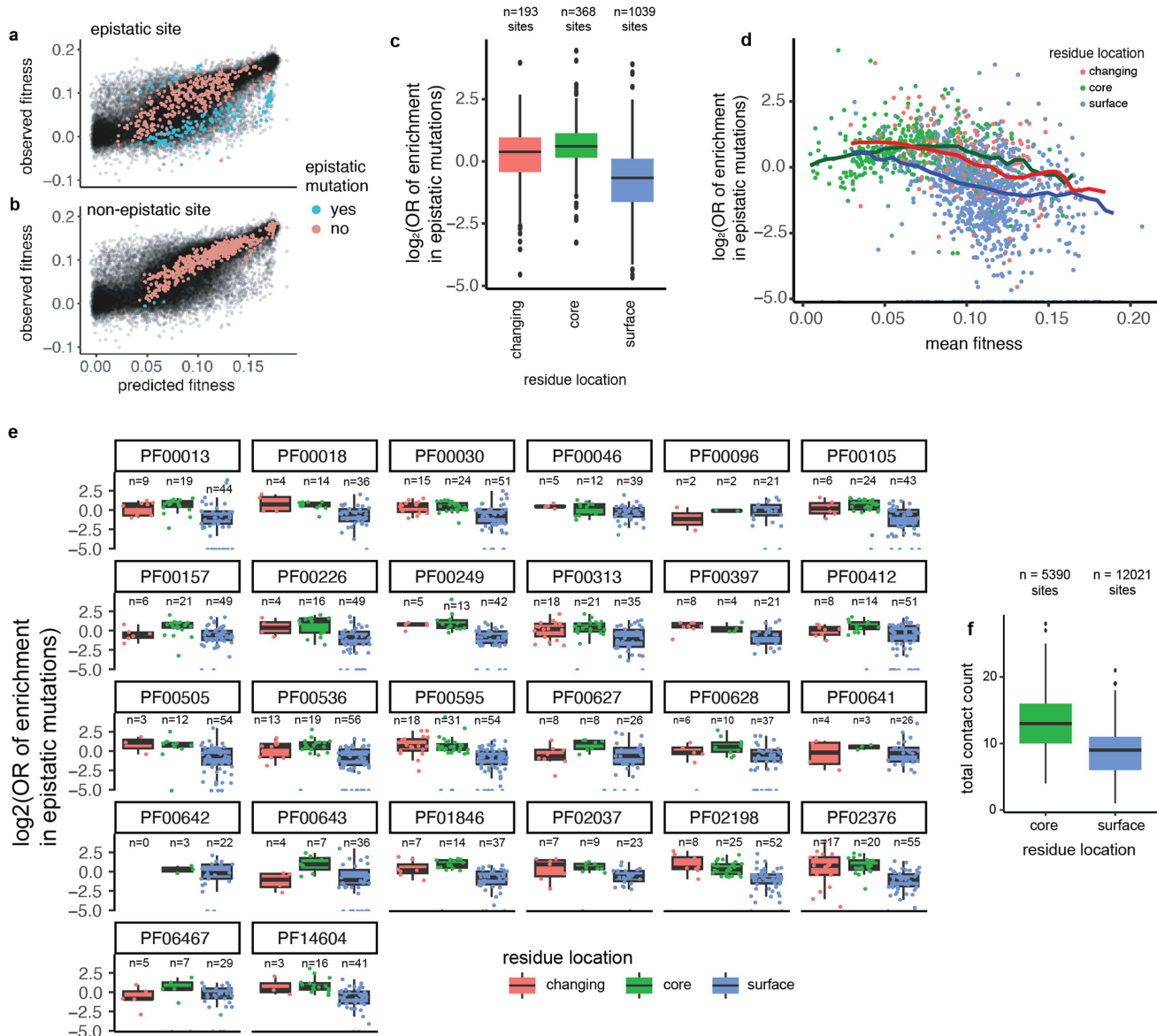


Extended Data Fig. 6 | Thermodynamic models of protein families.
a, Correlations between observed fitness values and energy model fitness predictions for test set variants held out in any of the 10 training folds, for

Boltzmann models across protein families. **b**, Inferred homologue-averaged folding $\Delta\Delta G$ s across protein families, normalized such that the 2.5th percentile of the data corresponds to a scaled $\Delta\Delta G$ of -1.



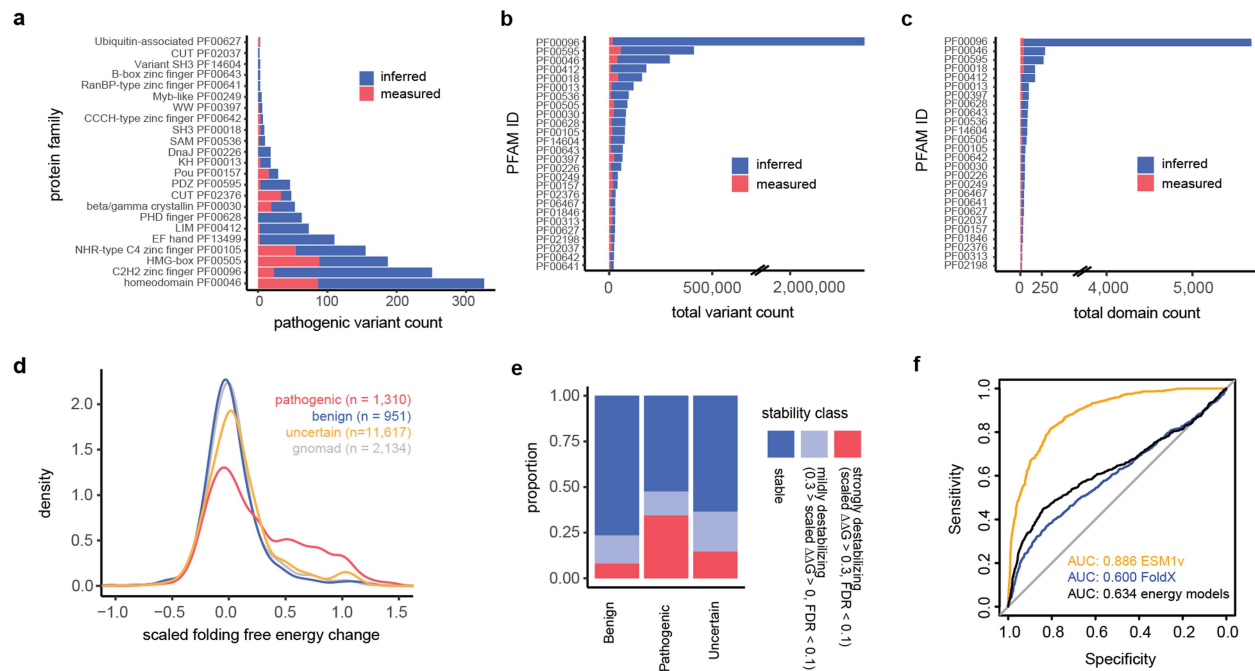
Extended Data Fig. 7 | Performance on left out protein domains. Correlation between model performance on left out protein domains, and the averaged genetic distance of the left-out domains to the rest of the training set, quantified as the hamming distance (families with $n > 9$ homologues). Pearson's r and its corresponding p -value are shown.



Extended Data Fig. 8 | Protein cores are more epistatic than surfaces.

a,b, Representative examples of an epistatic site (**a**) and a non-epistatic site (**b**) in the PDZ family. Epistatic mutations ($\text{FDR} < 0.1$, $|\text{residual}| > 0.05 \text{ h}^{-1}$) are shown in blue, and non-epistatic mutations are shown in pink. The epistatic site is enriched in epistatic mutations. **c**, Distributions of enrichment of epistatic mutations in protein core sites ($\text{rSASA} < 25\%$ in $> 75\%$ of homologues), surface sites ($\text{rSASA} > 25\%$ in $> 75\%$ of homologues), or changing sites. **d**, Enrichment of epistatic mutations in core ($n = 368$), surface ($n = 1,039$), and changing ($n = 193$) sites as a function of the average fitness of all variants per site. The running

median of each class of sites across the fitness range (window size = 0.05 h^{-1}) is shown as a coloured line. **e**, Distribution of enrichment of epistatic mutations in core, surface, and changing sites across protein families. **f**, Distribution of total contact numbers per residue in core ($n = 87,316$) and surface ($n = 193,797$) residues across the Domainome, estimated from getcontacts. Boxes in **c**, **e**, and **f** indicate the median, Q1 and Q3; the lower whisker indicates the lowest value no further than $Q1 - 1.5 \times \text{IQR}$; and the upper whisker indicates the largest value no further than $Q3 + 1.5 \times \text{IQR}$.



Extended Data Fig. 9 | Proteome-wide extension of stability predictions.
a, Number of pathogenic clinical variants covered by the predictions of homologue-averaged folding $\Delta\Delta G$ s. Colours represent whether variants are present in the aPCA fitness dataset (red) or not (blue). **b**, Number of measured and predicted protein variants, by family. **c**, Number of measured and predicted protein domains, by family. **d**, Distribution of scaled folding $\Delta\Delta G$ values of

pathogenic, benign, uncertain, and gnomAD variants (allele frequency $> 10^{-5}$). **e**, Proportion of mildly (one-tailed Z-test, $FDR < 0.1$, scaled folding $\Delta\Delta G > 0$) and strongly (one-tailed Z-test, $FDR < 0.1$, scaled folding $\Delta\Delta G > 0.3$) destabilizing variants across classes of clinical variants. **f**, ROC curves comparing clinical variant classification performance of ESM1v, FoldX, and energy models of protein families.

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R packages and versions:

```
[ 1] scales_1.2.1  msir_1.3.3  minpack.lm_1.2-2
[ 4] factoextra_1.0.7 drc_3.0-1  MASS_7.3-58.1
[ 7] dplyr_1.1.4    bio3d_2.4-3  bigsnpr_1.11.6
[10] bigstatsr_1.5.12 OneR_2.2    ggrepel_0.9.1
[13] reshape2_1.4.4 mltools_0.3.5 PRROC_1.3.1
[16] pROC_1.18.5  stringr_1.5.1  ggpubr_0.4.0
[19] viridis_0.6.2  viridisLite_0.4.1 GGally_2.1.2
[22] ggplot2_3.4.4  data.table_1.14.2
```

Other software:

Foldseek: <https://github.com/steineggerlab/foldseek>

Gephi: <https://gephi.org/>

hmmer: <http://hmmer.org/>

esm1v: <https://github.com/facebookresearch/esm>

thermoMPNN: <https://github.com/Kuhlman-Lab/ThermoMPNN>

getcontacts: <https://github.com/getcontacts/getcontacts>

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

We have deposited the data in SRA/GEO with the accession number GSE265942.

Publicly available data sources for analysis:

- PFAM domain annotations and alignments: <https://www.ebi.ac.uk/interpro/download/Pfam/>
- AlphaFold2 structures: <https://alphafold.ebi.ac.uk/>
- Conserved Domains Database (CDD) annotations through the InterPro API: <https://github.com/ProteinsWebTeam/interpro7-api/tree/master/docs>
- gnomAD v4 variants: <https://gnomad.broadinstitute.org/>
- clinical variants: clinvar (<https://www.ncbi.nlm.nih.gov/clinvar/>) and UniProt (using proteinsAPI: <https://www.ebi.ac.uk/proteins/api/doc/>)

Publicly available precomputed variant predictions

- AlphaMissense: https://console.cloud.google.com/storage/browser/dm_alphamissense
- EVE/popEVE/Tranception scores: <https://pop.evemodel.org/>
- FoldX: <https://ftp.ebi.ac.uk/pub/databases/ProtVar/predictions/stability>
- RaSP: <https://erda.ku.dk/archives/338b8e4a2a6124883ccd9f0fba9d37cc/published-archive.html>

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender

NA

Reporting on race, ethnicity, or other socially relevant groupings

NA

Population characteristics

NA

Recruitment

NA

Ethics oversight

NA

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences

☐ Behavioural & social sciences

☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see [nature.com/documents/nr-reporting-summary-flat.pdf](https://www.nature.com/documents/nr-reporting-summary-flat.pdf)

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size

We decided to do 500-1000 protein domains as this was as many we could assay with our resources. We did not calculate in advance what sample size was required for the downstream analyses.

Data exclusions

Data filtering and normalization

We computed several quality control metrics on a per-domain basis. First, the wild-type position in the fitness distribution, as the difference between the wild-type and the 95th percentile of the distribution, divided by the difference between the 95th and 5th percentile in the distribution (fitness 90% range). Second, the Pearson's correlation coefficient between replicates for missense variants. Third, the correlation

between fitness and solvent accessibility for all variants in each domain. And fourth, the correlation between fitness and hydrophobicity of all variants in each domain (measured as the first principal component of a comprehensive table of amino acid properties⁶⁸).

Compared to folded domains (more than 10% core residues and pLDDT>50), disordered and domains without a well-defined hydrophobic core (see Library Design) behave in a very distinct fashion in aPCA, with the wild type located in the middle of the fitness distribution, hydrophobicity negatively correlated with protein abundance, and narrow dynamic ranges resulting from typically small effects of mutations (Extended Data Fig. 1c). We thus excluded these domains from further analysis in the main text. Folded domains showed larger dynamic ranges, with the wild-type among the fittest variants, a negative correlation between mutation sensitivity and solvent accessibility, and a lack of correlation to hydrophobicity (Extended Data Fig 1c). For folded domains, the four quality metrics described above were highly correlated with each other, and we combined them using principal component analysis. The first principal component explained 64% of the variance, and was used as a single quality metric to rank all domains.

We retained all domains ranked 600 or higher, with a missense Pearson's $r > 0.485$, and with more than 50% of variants measured (with at least 10 counts in at least 1 replicate), resulting in 538 domains. We removed from this set 16 additional domains that were not compatible with 2-state folding, either showing many large effect abundance-increasing mutations (O75364_PFO0046_64, O75956_PFO9806_73, P10242_PFO0249_89, P52952_PFO0046_140, Q13263_PFO0643_205, Q86TZ1_PF13181_60, Q8IX03_PFO0397_1, Q8NDW8_PF13181_799, Q9Y2H9_PF17820_968, Q9Y6M9_PFO5347_15), narrow fitness ranges (P35637_PFO0641_421, EHEE-rd2-0005_1, HHH-rd2-0133_1), or clear correlations to hydrophobicity (E9PAV3_PF19026_2040, HEEH-rd3-0223_1, Q5VTD9_PFO0096_193), resulting in a final set of 522 domains.

The distributions of quality metrics for retained and discarded folded domains, and for disordered domains and folded domains without a well-defined core are in Extended Data Fig. 1c. Of the 478 discarded folded domains, 224 had slow growth rates as wt (< 0.075 h⁻¹), 176 had an unfit wild type (wt position < 0.4), 107 were negatively correlated with hydrophobicity (Pearson's $r < -0.25$), and 158 were not correlated with solvent accessibility (Pearson's $r < 0$) (we note that many of the discarded domains meet more than one of these criteria).

Replication	Triplicate. All attempts at replication were successful.
Randomization	NA. All mutations are measured at once, which effectively is blind and random. The calculations are automated by code and done blindly.
Blinding	NA. All mutations are measured at once, which effectively is blind and random. The calculations are automated by code and done blindly.

Behavioural & social sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description	Briefly describe the study type including whether data are quantitative, qualitative, or mixed-methods (e.g. qualitative cross-sectional, quantitative experimental, mixed-methods case study).
Research sample	State the research sample (e.g. Harvard university undergraduates, villagers in rural India) and provide relevant demographic information (e.g. age, sex) and indicate whether the sample is representative. Provide a rationale for the study sample chosen. For studies involving existing datasets, please describe the dataset and source.
Sampling strategy	Describe the sampling procedure (e.g. random, snowball, stratified, convenience). Describe the statistical methods that were used to predetermine sample size OR if no sample-size calculation was performed, describe how sample sizes were chosen and provide a rationale for why these sample sizes are sufficient. For qualitative data, please indicate whether data saturation was considered, and what criteria were used to decide that no further sampling was needed.
Data collection	Provide details about the data collection procedure, including the instruments or devices used to record the data (e.g. pen and paper, computer, eye tracker, video or audio equipment) whether anyone was present besides the participant(s) and the researcher, and whether the researcher was blind to experimental condition and/or the study hypothesis during data collection.
Timing	Indicate the start and stop dates of data collection. If there is a gap between collection periods, state the dates for each sample cohort.
Data exclusions	If no data were excluded from the analyses, state so OR if data were excluded, provide the exact number of exclusions and the rationale behind them, indicating whether exclusion criteria were pre-established.
Non-participation	State how many participants dropped out/declined participation and the reason(s) given OR provide response rate OR state that no participants dropped out/declined participation.
Randomization	If participants were not allocated into experimental groups, state so OR describe how participants were allocated to groups, and if allocation was not random, describe how covariates were controlled.

Ecological, evolutionary & environmental sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description	Briefly describe the study. For quantitative data include treatment factors and interactions, design structure (e.g. factorial, nested, hierarchical), nature and number of experimental units and replicates.
Research sample	Describe the research sample (e.g. a group of tagged <i>Passer domesticus</i> , all <i>Stenocereus thurberi</i> within Organ Pipe Cactus National

Research sample	<i>Monument), and provide a rationale for the sample choice. When relevant, describe the organism taxa, source, sex, age range and any manipulations. State what population the sample is meant to represent when applicable. For studies involving existing datasets, describe the data and its source.</i>
Sampling strategy	<i>Note the sampling procedure. Describe the statistical methods that were used to predetermine sample size OR if no sample-size calculation was performed, describe how sample sizes were chosen and provide a rationale for why these sample sizes are sufficient.</i>
Data collection	<i>Describe the data collection procedure, including who recorded the data and how.</i>
Timing and spatial scale	<i>Indicate the start and stop dates of data collection, noting the frequency and periodicity of sampling and providing a rationale for these choices. If there is a gap between collection periods, state the dates for each sample cohort. Specify the spatial scale from which the data are taken</i>
Data exclusions	<i>If no data were excluded from the analyses, state so OR if data were excluded, describe the exclusions and the rationale behind them, indicating whether exclusion criteria were pre-established.</i>
Reproducibility	<i>Describe the measures taken to verify the reproducibility of experimental findings. For each experiment, note whether any attempts to repeat the experiment failed OR state that all attempts to repeat the experiment were successful.</i>
Randomization	<i>Describe how samples/organisms/participants were allocated into groups. If allocation was not random, describe how covariates were controlled. If this is not relevant to your study, explain why.</i>
Blinding	<i>Describe the extent of blinding used during data acquisition and analysis. If blinding was not possible, describe why OR explain why blinding was not relevant to your study.</i>

Did the study involve field work? ☐ Yes ☐ No

Field work, collection and transport

Field conditions	<i>Describe the study conditions for field work, providing relevant parameters (e.g. temperature, rainfall).</i>
Location	<i>State the location of the sampling or experiment, providing relevant parameters (e.g. latitude and longitude, elevation, water depth).</i>
Access & import/export	<i>Describe the efforts you have made to access habitats and to collect and import/export your samples in a responsible manner and in compliance with local, national and international laws, noting any permits that were obtained (give the name of the issuing authority, the date of issue, and any identifying information).</i>
Disturbance	<i>Describe any disturbance caused by the study and how it was minimized.</i>

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used	<i>Describe all antibodies used in the study; as applicable, provide supplier name, catalog number, clone name, and lot number.</i>
Validation	<i>Describe the validation of each primary antibody for the species and application, noting any validation statements on the manufacturer's website, relevant citations, antibody profiles in online databases, or data provided in the manuscript.</i>

Eukaryotic cell lines

Policy information about [cell lines and Sex and Gender in Research](#)

Cell line source(s)	State the source of each cell line used and the sex of all primary cell lines and cells derived from human participants or vertebrate models.
Authentication	Describe the authentication procedures for each cell line used OR declare that none of the cell lines used were authenticated.
Mycoplasma contamination	Confirm that all cell lines tested negative for mycoplasma contamination OR describe the results of the testing for mycoplasma contamination OR declare that the cell lines were not tested for mycoplasma contamination.
Commonly misidentified lines (See ICLAC register)	Name any commonly misidentified cell lines used in the study and provide a rationale for their use.

Palaeontology and Archaeology

Specimen provenance	Provide provenance information for specimens and describe permits that were obtained for the work (including the name of the issuing authority, the date of issue, and any identifying information). Permits should encompass collection and, where applicable, export.
Specimen deposition	Indicate where the specimens have been deposited to permit free access by other researchers.
Dating methods	If new dates are provided, describe how they were obtained (e.g. collection, storage, sample pretreatment and measurement), where they were obtained (i.e. lab name), the calibration program and the protocol for quality assurance OR state that no new dates are provided.
☐ Tick this box to confirm that the raw and calibrated dates are available in the paper or in Supplementary Information.	
Ethics oversight	Identify the organization(s) that approved or provided guidance on the study protocol, OR state that no ethical approval or guidance was required and explain why not.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Animals and other research organisms

Policy information about [studies involving animals; ARRIVE guidelines](#) recommended for reporting animal research, and [Sex and Gender in Research](#)

Laboratory animals	For laboratory animals, report species, strain and age OR state that the study did not involve laboratory animals.
Wild animals	Provide details on animals observed in or captured in the field; report species and age where possible. Describe how animals were caught and transported and what happened to captive animals after the study (if killed, explain why and describe method; if released, say where and when) OR state that the study did not involve wild animals.
Reporting on sex	Indicate if findings apply to only one sex; describe whether sex was considered in study design, methods used for assigning sex. Provide data disaggregated for sex where this information has been collected in the source data as appropriate; provide overall numbers in this Reporting Summary. Please state if this information has not been collected. Report sex-based analyses where performed, justify reasons for lack of sex-based analysis.
Field-collected samples	For laboratory work with field-collected samples, describe all relevant parameters such as housing, maintenance, temperature, photoperiod and end-of-experiment protocol OR state that the study did not involve samples collected from the field.
Ethics oversight	Identify the organization(s) that approved or provided guidance on the study protocol, OR state that no ethical approval or guidance was required and explain why not.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Clinical data

Policy information about [clinical studies](#)

All manuscripts should comply with the ICMJE [guidelines for publication of clinical research](#) and a completed [CONSORT checklist](#) must be included with all submissions.

Clinical trial registration	Provide the trial registration number from ClinicalTrials.gov or an equivalent agency.
Study protocol	Note where the full trial protocol can be accessed OR if not available, explain why.
Data collection	Describe the settings and locales of data collection, noting the time periods of recruitment and data collection.

Outcomes

Describe how you pre-defined primary and secondary outcome measures and how you assessed these measures.

Dual use research of concern

Policy information about [dual use research of concern](#)

Hazards

Could the accidental, deliberate or reckless misuse of agents or technologies generated in the work, or the application of information presented in the manuscript, pose a threat to:

No	Yes
☐	☐ Public health
☐	☐ National security
☐	☐ Crops and/or livestock
☐	☐ Ecosystems
☐	☐ Any other significant area

Experiments of concern

Does the work involve any of these experiments of concern:

No	Yes
☐	☐ Demonstrate how to render a vaccine ineffective
☐	☐ Confer resistance to therapeutically useful antibiotics or antiviral agents
☐	☐ Enhance the virulence of a pathogen or render a nonpathogen virulent
☐	☐ Increase transmissibility of a pathogen
☐	☐ Alter the host range of a pathogen
☐	☐ Enable evasion of diagnostic/detection modalities
☐	☐ Enable the weaponization of a biological agent or toxin
☐	☐ Any other potentially harmful combination of experiments and agents

Plants

Seed stocks

Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.

Novel plant genotypes

Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.

Authentication

Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosaicism, off-target gene editing) were examined.

ChIP-seq

Data deposition

- ☐ Confirm that both raw and final processed data have been deposited in a public database such as [GEO](#).
- ☐ Confirm that you have deposited or provided access to graph files (e.g. BED files) for the called peaks.

Data access links

May remain private before publication.

For "Initial submission" or "Revised version" documents, provide reviewer access links. For your "Final submission" document, provide a link to the deposited data.

Files in database submission

Provide a list of all files available in the database submission.

Genome browser session (e.g. [UCSC](#))

Provide a link to an anonymized genome browser session for "Initial submission" and "Revised version" documents only, to enable peer review. Write "no longer applicable" for "Final submission" documents.

Methodology

Replicates	<i>Describe the experimental replicates, specifying number, type and replicate agreement.</i>
Sequencing depth	<i>Describe the sequencing depth for each experiment, providing the total number of reads, uniquely mapped reads, length of reads and whether they were paired- or single-end.</i>
Antibodies	<i>Describe the antibodies used for the ChIP-seq experiments; as applicable, provide supplier name, catalog number, clone name, and lot number.</i>
Peak calling parameters	<i>Specify the command line program and parameters used for read mapping and peak calling, including the ChIP, control and index files used.</i>
Data quality	<i>Describe the methods used to ensure data quality in full detail, including how many peaks are at FDR 5% and above 5-fold enrichment.</i>
Software	<i>Describe the software used to collect and analyze the ChIP-seq data. For custom code that has been deposited into a community repository, provide accession details.</i>

Flow Cytometry

Plots

Confirm that:

- ☐ The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).
- ☐ The axis scales are clearly visible. Include numbers along axes only for bottom left plot of group (a 'group' is an analysis of identical markers).
- ☐ All plots are contour plots with outliers or pseudocolor plots.
- ☐ A numerical value for number of cells or percentage (with statistics) is provided.

Methodology

Sample preparation	<i>Describe the sample preparation, detailing the biological source of the cells and any tissue processing steps used.</i>
Instrument	<i>Identify the instrument used for data collection, specifying make and model number.</i>
Software	<i>Describe the software used to collect and analyze the flow cytometry data. For custom code that has been deposited into a community repository, provide accession details.</i>
Cell population abundance	<i>Describe the abundance of the relevant cell populations within post-sort fractions, providing details on the purity of the samples and how it was determined.</i>
Gating strategy	<i>Describe the gating strategy used for all relevant experiments, specifying the preliminary FSC/SSC gates of the starting cell population, indicating where boundaries between "positive" and "negative" staining cell populations are defined.</i>
☐ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.	

Magnetic resonance imaging

Experimental design

Design type	<i>Indicate task or resting state; event-related or block design.</i>
Design specifications	<i>Specify the number of blocks, trials or experimental units per session and/or subject, and specify the length of each trial or block (if trials are blocked) and interval between trials.</i>
Behavioral performance measures	<i>State number and/or type of variables recorded (e.g. correct button press, response time) and what statistics were used to establish that the subjects were performing the task as expected (e.g. mean, range, and/or standard deviation across subjects).</i>

Acquisition

Imaging type(s)	<i>Specify: functional, structural, diffusion, perfusion.</i>
Field strength	<i>Specify in Tesla</i>
Sequence & imaging parameters	<i>Specify the pulse sequence type (gradient echo, spin echo, etc.), imaging type (EPI, spiral, etc.), field of view, matrix size, slice thickness, orientation and TE/TR/flip angle.</i>
Area of acquisition	<i>State whether a whole brain scan was used OR define the area of acquisition, describing how the region was determined.</i>
Diffusion MRI	☐ Used ☐ Not used

Preprocessing

Preprocessing software	<i>Provide detail on software version and revision number and on specific parameters (model/functions, brain extraction, segmentation, smoothing kernel size, etc.).</i>
Normalization	<i>If data were normalized/standardized, describe the approach(es): specify linear or non-linear and define image types used for transformation OR indicate that data were not normalized and explain rationale for lack of normalization.</i>
Normalization template	<i>Describe the template used for normalization/transformation, specifying subject space or group standardized space (e.g. original Talairach, MNI305, ICBM152) OR indicate that the data were not normalized.</i>
Noise and artifact removal	<i>Describe your procedure(s) for artifact and structured noise removal, specifying motion parameters, tissue signals and physiological signals (heart rate, respiration).</i>
Volume censoring	<i>Define your software and/or method and criteria for volume censoring, and state the extent of such censoring.</i>

Statistical modeling & inference

Model type and settings	<i>Specify type (mass univariate, multivariate, RSA, predictive, etc.) and describe essential details of the model at the first and second levels (e.g. fixed, random or mixed effects; drift or auto-correlation).</i>
Effect(s) tested	<i>Define precise effect in terms of the task or stimulus conditions instead of psychological concepts and indicate whether ANOVA or factorial designs were used.</i>
Specify type of analysis:	☐ Whole brain ☐ ROI-based ☐ Both
Statistic type for inference	<i>Specify voxel-wise or cluster-wise and report all relevant parameters for cluster-wise methods.</i>
(See Eklund et al. 2016)	
Correction	<i>Describe the type of correction and how it is obtained for multiple comparisons (e.g. FWE, FDR, permutation or Monte Carlo).</i>

Models & analysis

n/a	Involved in the study
☐	☐ Functional and/or effective connectivity
☐	☐ Graph analysis
☐	☐ Multivariate modeling or predictive analysis
Functional and/or effective connectivity	<i>Report the measures of dependence used and the model details (e.g. Pearson correlation, partial correlation, mutual information).</i>
Graph analysis	<i>Report the dependent variable and connectivity measure, specifying weighted graph or binarized graph, subject- or group-level, and the global and/or node summaries used (e.g. clustering coefficient, efficiency, etc.).</i>
Multivariate modeling and predictive analysis	<i>Specify independent variables, features extraction and dimension reduction, model, training and evaluation metrics.</i>